

A Cramér-von Mises-type test statistic for identifying jump variations in high-frequency time series

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Abstract

This paper studies a test statistic of the Cramér-Von Mises type for detecting jumps in addition to a continuous local martingale, based on the empirical distribution of devolatilized increments. The devolatilized increments are properly formed by resizing and truncating the gross increments in order to remove the effects of stochastic volatility and big jumps. The main result and contribution of this study shows that the distribution function of the proposed test statistic follows approximately a gamma distribution. This is of vital importance because it enables us to determine the critical region for the rejection of the null hypothesis of interest. We observe better results in comparison with the Kolmogorov-Smirnov (KS) test. Specifically, we show that the power and the error rate of the test using Cramér-von Mises (Cv-M) statistic is better than those using the KS test statistic, showing a higher detection power and lower error rate. We applied the proposed test to three real data sets, namely, the stock returns of Google, Apple and Goldman Sachs (GS), and found that the proposed test can capture the dynamics of the series.

Key words: Itô semimartingale, Activity Jumps, Variation Jumps, Distribution function, Cramér-von Mises.

JEL Codes: C10, C14, C15, C55, C58.

1 Introduction

Stochastic models based on diffusions are often used to describe the complex dynamical systems in biology, engineering, finance, and physics. However, these models, when applied in finance, do not take into account possible price jumps during a business session on a stock exchange due to the

arrival of market information. In diffusion models, price movements are conditionally Gaussian, so large and sudden movements do not occur. On the other hand, in practice, price jumps can give rise to substantial losses or gains.

The volatility of financial time series has been treated in two ways. The first approach is parametric and postulates a latent variable via conditional heteroscedastic models to describe the volatility, such as the ARCH (Autoregressive Conditional Heteroscedasticity) family models proposed by Engle (1982) and the stochastic volatility models. There is a huge literature on this approach and for details, see, for example, Tsay (2005) or Morettin (2017). This parametric approach is used mainly for low frequency data (daily, weekly, monthly) of financial asset returns. With the availability of intraday data (also called high-frequency data), a second approach, which is nonparametric, was considered. The main motivation for using the nonparametric approach is because parametric models often fail to adequately capture the movements of intraday volatility (see Andersen et al., 2003).

A nonparametric approach consists of constructing the daily realized volatility (RV) of a financial series using intraday returns, sampled at intervals Δt , of the order of 5 or 15 minutes, for example. RV is obtained from sums of squares of intraday returns (see Andersen et al., 2003 and Barndorff-Nielsen and Shephard, 2002). Another method is to construct the realized bi-power variance (RVBP), from the sums of cross products of absolute adjacent, properly scaled returns (Barndorff-Nielsen and Shephard, 2006). For details, see Aït-Sahalia and Jacob (2014).

Fan and Wang (2008), Barndorff-Nielsen and Shephard (2006) and Aït-Sahalia and Jacob (2009) derived test statistics for detecting the existence of jumps. Others authors established tests for the necessity of adding a Brownian force, see Aït-Sahalia and Jacob (2010), Jing et al. (2012), Kong et al. (2015), Todorov and Tauchen (2014) and Todorov (2015). Aït-Sahalia and Jacob (2011) studied whether the jump component is of finite activity when the Brownian force is present. Recently, Kong (2017) studied whether it is necessary to add an infinite variation jump term in addition to a continuous local martingale, using a Kolmogorov-Smirnov type test statistic.

In recent years, marked efforts have been devoted to study the distribution function of the jump test statistics. Csörgö and Faraway (1996) showed the exact and asymptotic distribution of Cramér-von Mises statistic when the empirical distribution function is a uniform distribution function. Todorov and Tauchen (2014) suggested using other measures of discrepancy between distributions like the Cramér-von Mises test for the presence of diffusion component in X_t .

However, finding the distribution function of a jump test statistic is not easy so that numerical methods are often employed. In this study, we propose a new test statistic of the modified Cramér-von Mises type. Our simulation study showed that the proposed test statistic follows approximately a known distribution function. Comparing the respective quantiles at the significance levels of 5% and 10%, the values of the distribution found and the sample distribution are very close.

The rest of the paper is organized as follows. We state the models considered in Section 2 and present the empirical cumulative distribution function (c.d.f.) and estimators for volatility in Section 3. We then evaluate the performance of the empirical c.d.f. in Section 4 and discuss the choice of tuning parameters in Section 5. Section 6 provides the quantile analysis whereas Section 7 presents the results of a simulation study for the proposed test statistic. Section 8 applies the proposed test statistic to real data.

2 Setup

Let $(\Omega, \mathcal{F}_t, \mathcal{F}, P)$ be a filtered probability space. The standard jump-diffusion model used for modeling many stochastic processes is an Itô semimartingale given by the following differential equation:

$$dX_t = \alpha_t dt + \sigma_t dW_t + dZ_t, \quad (1)$$

where α_t and σ_t are processes with Cadlag paths, W_t is a standard Brownian motion, and Z_t is an Itô semimartingale process of pure-jump type. Todorov and Tauchen (2014) provide a formal setup and assumptions. They generalize the setup (1) to accommodate the alternative hypothesis in which X_t can be of pure-jump type. Itô semimartingale plays an important role in stochastic calculus and we make the following assumption:

Assumption 1. *Suppose that Y_t follows a non-parametric volatility model:*

$$Y_t = X_t + \epsilon_t, \quad t \in [0, 1]. \quad (2)$$

in which X_t is a continuous Itô semimartingale, that is,

$$X_t = X_0 + \int_0^t b_s ds + \int_0^t \sigma_s dW_s + Z_t, \quad (3)$$

where $\int_0^t b_s ds$ is the drift term with b_s being an optional and Cadlag process, $\int_0^t \sigma_s dW_s$ is a continuous local martingale with σ_s being an adapted process, W_s is a standard Brownian motion, and Z_t is a skewed β -stable Lévy process.

Kong (2017) for the first time provided a theoretical test for the presence of infinite variation jumps in the simultaneous presence of a diffusion term and a jump component of finite variation and established the asymptotic theory of the empirical distribution of the "devolatilized" increments of Itô semimartingales with infinitely active or even infinite variation jumps. To estimate the spot volatility, he uses the local method and splits the interval into non-overlapping shrinking blocks with each block length equal to $2v_n$ consisting of $2k_n$ intervals of length Δ_n , where k_n is some integer depending on n . The local estimate of $\sigma_j^2 = \sigma_{2jk_n\Delta_n}$ is

$$\hat{\sigma}_j^2(u) = c_j(u) - \frac{1}{u^2 k_n} (\sinh(u^2 c_j(u)))^2, \quad (4)$$

where

$$\begin{aligned} c_j(u) &= -\frac{1}{u^2} \log \left(L_j(u) \vee \frac{c}{\sqrt{k_n}} \right), \quad 0 \leq j \leq \lfloor n/(2k_n) \rfloor - 1, \\ L_j(u) &= \frac{1}{k_n} \sum_{l=1}^{k_n} \cos \left(u \frac{\Delta_{2jk_n+2l}^n X - \Delta_{2jk_n+2l-1}^n X}{\sqrt{\Delta_n}} \right) \end{aligned}$$

where the subtracted term in (4) is used to correct the bias due to the jumps, $a \vee b = \max(a, b)$, the lower threshold $\frac{c}{\sqrt{k_n}}$ in $c_j(u)$ with constant c is to assure that the logarithmic function is well-defined, and $\lfloor m \rfloor$ denotes the integer part of m . There are other methods to estimate the spot volatility. See, for instance, Todorov and Tauchen (2014) and Fang and Wang (2007). The major advantage of this Laplace-transform-based local estimator is that it can easily separate the effect of the Brownian force and the stable-like driving force. For properly chosen m_n and u_n the empirical distribution function of the devolatilized increments is defined as

$$\hat{F}_n(u_n, \tau) = \frac{1}{\lfloor n/2k_n \rfloor m_n} \sum_{j=1}^{\lfloor n/2k_n \rfloor} \sum_{i=2jk_n+1}^{2jk_n+m_n} I \left(\frac{\Delta_i^n X}{\sqrt{\hat{\sigma}_j^2(u_n) \Delta_n}} \leq \tau \right), \quad (5)$$

for $\tau \in R$, for details, see Kong (2017). Furthermore, the author defines the empirical process as

$$\hat{Y}_n(\tau) = \sqrt{\lfloor n/2k_n \rfloor m_n} (\hat{F}_n(u_n, \tau) - \Phi(\tau)), \quad (6)$$

where $\Phi(\tau)$ denotes the c.d.f. of a standard normal random variable.

In this paper, we consider the following hypotheses

$$H_0 : \Delta_i^n X. \quad (\text{standard normal model})$$

$$H_1 : \Delta_i^n X. \quad (\text{jumps-diffusion model})$$

Along the paper, we assume that the available data set $\{X_{t_j}; 0 \leq j \leq n\}$ which are discretely sampled from X , and are equally spaced in the fixed interval $[0, T]$, i.e, $t_j = j\Delta_n$ with $\Delta_n = T/n$ for $0 \leq j \leq n$, and $\Delta_i^n X = X_{t_i} - X_{t_{i-1}}$ denote the i th one-step increment for $1 \leq i \leq n$.

Our aim is to show that the Cramer-Von Mises statistic has better properties in terms of size and power compared to the Kolmogorov Smirnov test. To this end, we define the empirical process as:

$$\hat{P}_n(\tau) = \sqrt{\lfloor n/2k_n \rfloor m_n} (\hat{\Phi}_n(\tau) - \Phi(\tau)), \quad (7)$$

where $\hat{\Phi}_n(\tau)$ is an empirical distribution function of the devolatilized increments. In this paper we will consider two possible empirical distribution functions in (7).

Theorem 4 of Todorov and Tauchen (2014) and Theorem 1 of Kong (2017) motivate us to propose a measure of discrepancy between distributions using the "Cramér-Von Mises" type statistic as

$$T_A^n = d^2 \sqrt{\lfloor n/2k_n \rfloor m_n}, \quad (8)$$

where A is a compact set in R and

$$d^2 = \int_{-\infty}^{\infty} [F_n(x) - F^*(x)]^2 dF^*(x).$$

Let $F_n(x) = \hat{F}'_n(\tau)$ and $F^*(x) = \Phi(\tau)$, then

$$d^2 = \int_{-\infty}^{\infty} [\hat{F}'_n(\tau) - \Phi(\tau)]^2 d\Phi(\tau).$$

The choice for this test statistic comes from the fact that the statistic has better power than the KS statistic that measures the maximum distance. The critical region of the proposed test is given by

$$C_n = \left\{ d^2 \sqrt{[n/2k_n] m_n} > q_n(\alpha, A) \right\},$$

where $q_n(\alpha, A)$ is the $(1 - \alpha)$ -quantile of

$$\sqrt{[n/(2k_n)] m_n} \int_{-\infty}^{\infty} [\hat{F}'_n(\tau) - \Phi(\tau)]^2 d\Phi(\tau). \quad (9)$$

We evaluate $q_n(\alpha, A)$ via simulation. If $T_A^n > q_n(\alpha, A)$, we reject H_0 .

3 Empirical c.d.f. and Estimators for Volatility

We used the local estimator proposed by Todorov and Tauchen (2014) to estimate σ_t^2 . On each of the blocks the local estimator of σ_t^2 is given by

$$\hat{V}_j^n = \frac{\pi}{2} \frac{n}{k_n - 1} \sum_{i=(j-1)k_n+2}^{jk_n} |\Delta_{i-1}^n X| |\Delta_i^n X|, \quad j = 1, \dots, [n/2k_n], \quad (10)$$

which is the bipower variation for measuring the quadratic variation of the diffusion component of X . Todorov and Tauchen (2014) remove the high-frequency increments that contain "big" jumps. The total number of increments used in their statistic is thus given by

$$N^n(\alpha, \bar{w}) = \sum_{j=1}^{[n/2k_n]} \sum_{i=(j-1)k_n+1}^{(j-1)k_n+m_n} I \left(|\Delta_i^n X| \leq \alpha \sqrt{\hat{V}_j^n n^{\bar{w}}} \right),$$

where $\alpha > 0$ and $\bar{w} \in (0, 1/2)$. They use a time-varying threshold in the truncation to account for the time varying σ_t . The scaling of every high-frequency increment is done after adjusting $\hat{V}_j^n$ to exclude the contribution of that increment in its formation:

$$\hat{V}_j^n(i) = \begin{cases} \frac{k_n-1}{k_n-3} \hat{V}_j^n - \frac{\pi}{2} \frac{n}{k_n-3} |\Delta_i^n X| |\Delta_{i+1}^n X|, & \text{for } i = (j-1)k_n + 1, \\ \frac{k_n-1}{k_n-3} \hat{V}_j^n - \frac{\pi}{2} \frac{n}{k_n-3} (|\Delta_{i-1}^n X| |\Delta_i^n X| + |\Delta_i^n X| |\Delta_{i+1}^n X|), & \text{for } i = (j-1)k_n + 2, \dots, jk_n - 1 \\ \frac{k_n-1}{k_n-3} \hat{V}_j^n - \frac{\pi}{2} \frac{n}{k_n-3} |\Delta_{i-1}^n X| |\Delta_i^n X|, & \text{for } i = jk_n. \end{cases} \quad (11)$$

They then define

$$\hat{F}_n(\tau) = \frac{1}{N^n(\alpha, \bar{w})} \sum_{j=1}^{[n/2k_n]} \sum_{i=(j-1)k_n+1}^{(j-1)k_n+m_n} I \left(\frac{\sqrt{n} \Delta_i^n X}{\sqrt{\hat{V}_j^n(i)}} \leq \tau \right) I_{(|\Delta_i^n X| \leq \alpha \sqrt{\hat{V}_j^n n^{\bar{w}}})}, \quad (12)$$

which is simply the empirical c.d.f. of the devolatilized increments that do not contain any big jumps. In the jump-diffusion case of (1), $\hat{F}_n(\tau)$ should be approximately the c.d.f. of a standard normal random variable.

Todorov and Tauchen (2014) use an alternative estimator of the volatility that is the truncated variation defined as

$$\hat{C}_j^m = \frac{n}{k_n} \sum_{i=(j-1)k_n+1}^{jk_n} |\Delta_i^n X|^2 I(|\Delta_i^n X| \leq \alpha n^{\bar{w}}), \quad j = 1, \dots, \lfloor n/2k_n \rfloor,$$

where $\alpha > 0$ and $\bar{w} \in (0, 1/2)$ and the corresponding one excluding the contribution of the i th increment for $i = (j-1)k_n + 1, \dots, jk_n$, is

$$\hat{C}_j^m(i) = \frac{k_n}{k_n - 1} \hat{C}_j^m - \frac{n}{k_n - 1} |\Delta_i^n X|^2 I(|\Delta_i^n X| \leq \alpha n^{\bar{w}}), \quad j = 1, \dots, \lfloor n/2k_n \rfloor. \quad (13)$$

They also define the corresponding empirical c.d.f. of the devolatilized (and truncated) high-frequency increments as

$$\hat{F}'_n(\tau) = \frac{1}{N^n(\alpha, \bar{w})} \sum_{j=1}^{\lfloor n/k_n \rfloor} \sum_{i=(j-1)k_n+1}^{(j-1)k_n+m_n} I\left(\frac{\sqrt{n}\Delta_i^n X}{\sqrt{\hat{C}_j^m(i)}} \leq \tau\right) I_{(|\Delta_i^n X| \leq \alpha n^{\bar{w}})}, \quad (14)$$

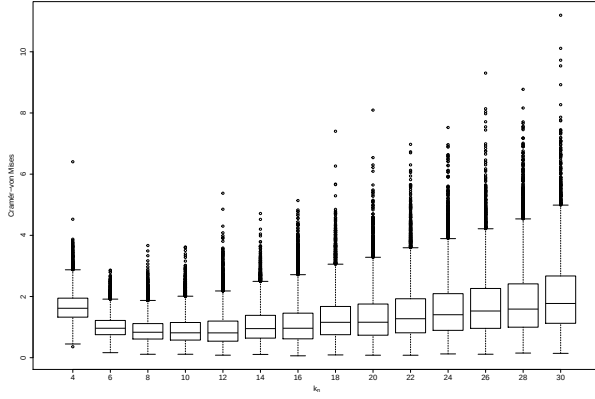
where $\alpha > 0$, $\bar{w} \in (0, 1/2)$ and

$$N^n(\alpha, \bar{w}) = \sum_{j=1}^{\lfloor n/k_n \rfloor} \sum_{i=(j-1)k_n+1}^{(j-1)k_n+m_n} I(|\Delta_i^n X| \leq \alpha n^{\bar{w}}).$$

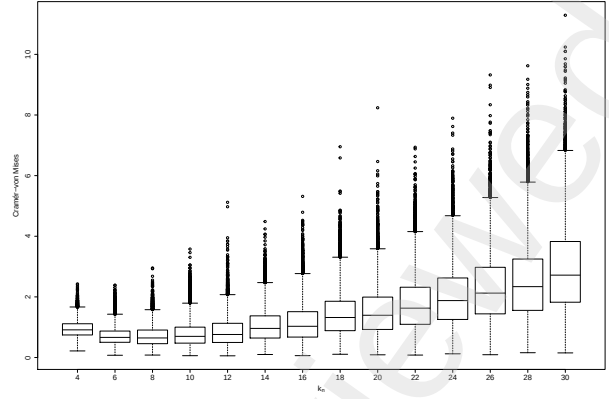
4 Performance of the empirical c.d.f.

We consider the cumulative distribution functions with their respective volatility estimators as described in (12) that contains the bipower variation estimator and in (14) that contains the truncated variation estimator. A simulation study was performed with 10,000 replications and different values of n , k_n and m_n . We set the pair of (k_n, m_n) for $n = 1000$ to be (4, 2), (6, 3), (8, 4), (10, 5), (12, 5), (14, 8), (16, 7), (18, 11), (20, 9), (22, 11), (24, 12), (26, 13), (28, 13), (30, 16). For $n = 2000$ to be (4, 2), (6, 3), (8, 4), (10, 5), (12, 5), (14, 8), (16, 7), (18, 11), (20, 9), (22, 11), (24, 12), (26, 13), (28, 13), (30, 16), (32, 15), (34, 18), (36, 17), (38, 20), (40, 20), (42, 26), (44, 21) and finally for $n = 5000$ we have the pairs (4, 2), (6, 3), (8, 4), (10, 5), (12, 5), (14, 8), (16, 7), (18, 11), (20, 9), (22, 11), (24, 12), (26, 13), (28, 13), (30, 16), (32, 15), (34, 18), (36, 17), (38, 20), (40, 20), (42, 26), (44, 21), (46, 25), (48, 23), (50, 28), (52, 27), (54, 34), (56, 31), (58, 30), (60, 34), (62, 33), (64, 30), (66, 40), (68, 35), (70, 33).

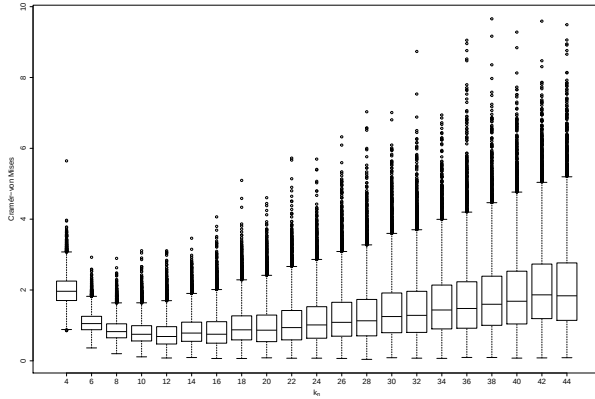
Figure 1 shows the behavior of the test statistic Cv-M against k_n for the jump-diffusion model. From the plots, we observe some outliers indicating the existence of some discrepant values. Also, the value of the test statistic increases as k_n increases. In cases where $k_n = 4$ and $k_n = 6$ the values of the test statistic present some inconsistencies. In these cases we simply call F the empirical c.d.f $\hat{F}_n(\tau)$ that contains the local estimator given in (11) and F' as the empirical c.d.f. $\hat{F}'_n(\tau)$ that contains the



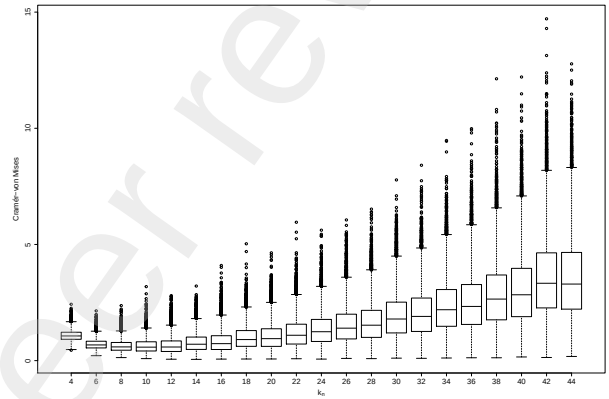
(a) Cv-M using F , $n = 1000$.



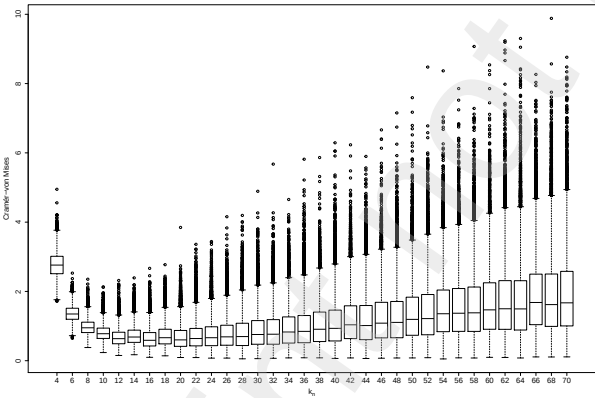
(b) Cv-M using F' , $n = 1000$.



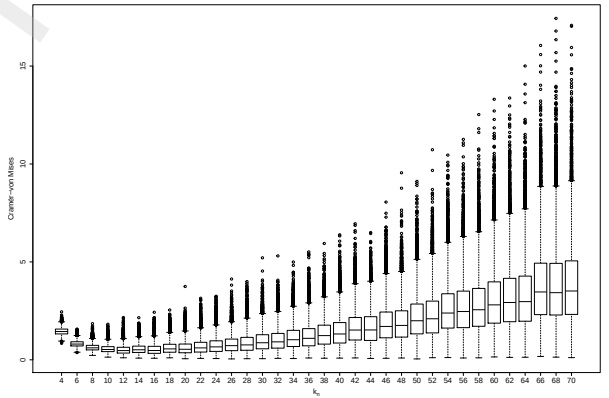
(c) Cv-M using F , $n = 2000$.



(d) Cv-M using F' , $n = 2000$.



(e) Cv-M using F , $n = 5000$.



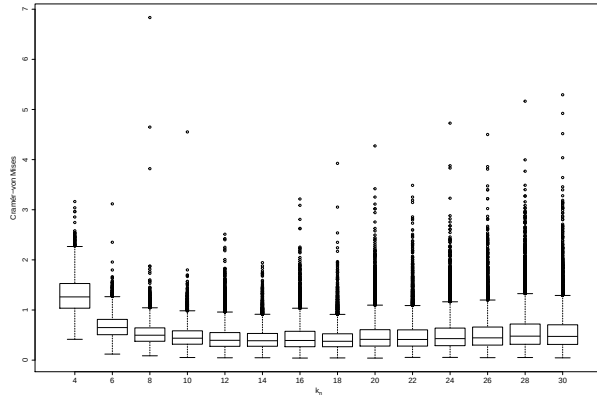
(f) Cv-M using F' , $n = 5000$.

Figure 1 – Left panel: the test statistic Cv-M against k_n using F ; Right panel: the test statistic Cv-M against k_n using F' for the jump diffusion model.

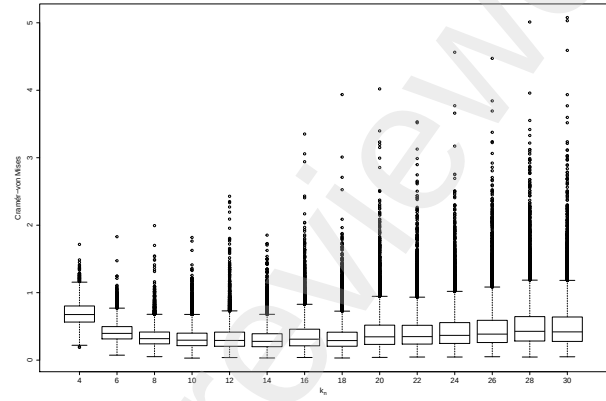
truncated variation estimator given in (13). Further, we call the test statistic using $\hat{F}_n(\tau)$ as $TA1$ and the test statistic using $\hat{F}'_n(\tau)$ as $TA2$.

Figure 2 shows the behavior of the test statistic Cv-M against k_n for the standard normal model.

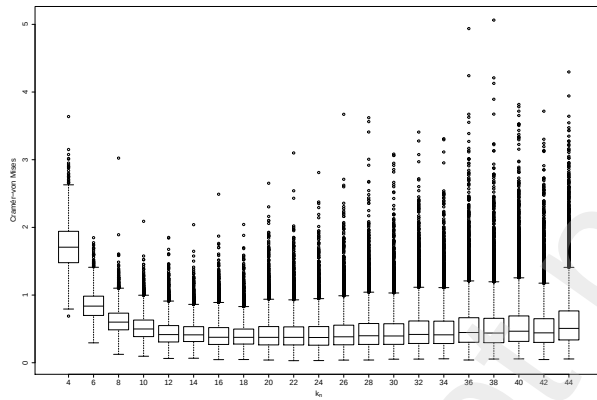
The first simulations were based on 10,000 replicas with $n = 1000$ and the same pairs of k_n



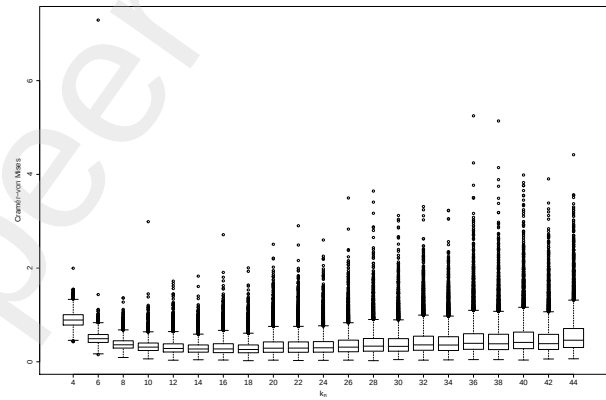
(a) Cv-M using F , $n = 1000$.



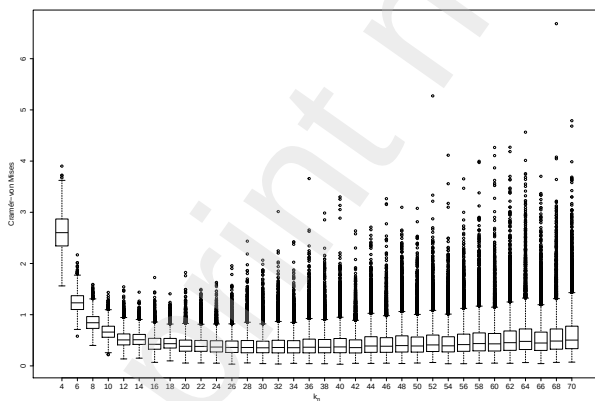
(b) Cv-M using F' , $n = 1000$.



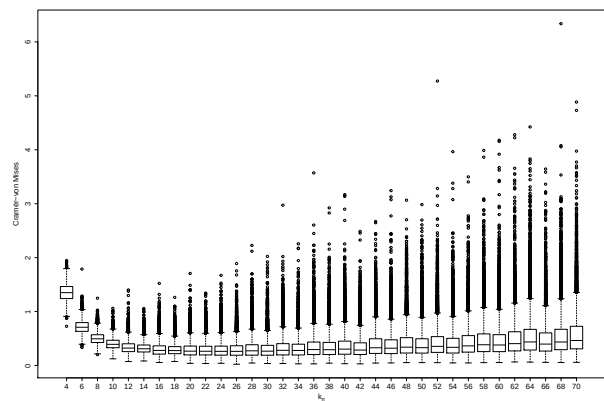
(c) Cv-M using F , $n = 2000$.



(d) Cv-M using F' , $n = 2000$.



(e) Cv-M using F , $n = 5000$.



(f) Cv-M using F' , $n = 5000$.

Figure 2 – Left panel: the test statistic Cv-M against k_n using F ; Right panel: the test statistic Cv-M against k_n using F' for the standard normal model.

and m_n as mentioned above. The idea is to observe which test statistic performs better using the two variance estimators mentioned in the previous section. It can be observed in Figure 3 that for the jump-diffusion model, the mean squared distance of the test statistic increases as k_n increases. On the other hand, for the normal model, on the average, the quadratic distance of the test statistic stabilizes, that is, the values of the test statistics are close as k_n increases. We present a method for choosing

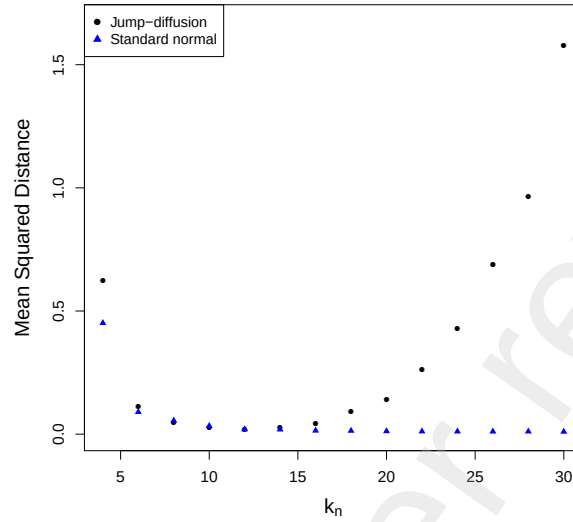


Figure 3 – Mean Squared Distance of the two test statistics for different values of k_n , in data generated from the two models with $n = 1000$.

the "best" critical value. The ROC curve (Receiver Operating Characteristic Curve) is widely used to determine the cutoff point. We test the critical value " q " between 1 and 3 and we use the ROC curve to show the cutoff point. We compare detection and error rates for this value and a fixed value.

Table 1 – Detection and error rates for k_n values between 4 and 30 for fixed critical value $q = 1.5$ and simulated the critical value $q = 1.2322$, in 10000 replicas of data generated with $n = 1000$.

k_n	m_n	$q = 1.5$				$q = 1.2322$			
		Error rate		Detection rate		Error rate		Detection rate	
		$TA1$	$TA2$	$TA1$	$TA2$	$TA1$	$TA2$	$TA1$	$TA2$
4	2	0.2728	0.0002	0.6026	0.0376	0.5286	0.0052	0.8217	0.1453
6	3	0.0029	0.0001	0.0952	0.0193	0.0204	0.0004	0.2378	0.0604
8	4	0.0015	0.0004	0.0805	0.0330	0.0056	0.0010	0.1793	0.0863
10	5	0.0011	0.0003	0.1061	0.0643	0.0057	0.0008	0.2074	0.1402
12	5	0.0042	0.0020	0.1372	0.1088	0.0100	0.0052	0.2324	0.1993
14	8	0.0014	0.0006	0.2071	0.1982	0.0054	0.0028	0.3248	0.3242
16	7	0.0060	0.0040	0.2323	0.2551	0.0164	0.0113	0.3490	0.3833
18	11	0.0039	0.0030	0.3204	0.4066	0.0110	0.0082	0.4578	0.5502
20	9	0.0140	0.0101	0.3381	0.4465	0.0287	0.0213	0.4616	0.5796
22	11	0.0149	0.0113	0.3984	0.5568	0.0306	0.0247	0.5208	0.6814
24	12	0.0164	0.0125	0.4569	0.6501	0.0354	0.0258	0.5795	0.7586
26	13	0.0231	0.0198	0.5094	0.7286	0.0451	0.0378	0.6255	0.8170
28	13	0.0309	0.0258	0.5378	0.7675	0.0585	0.0482	0.6472	0.8477
30	16	0.0305	0.0272	0.5989	0.8390	0.0599	0.0500	0.7057	0.8937

Table 1 shows that for a fixed critical value 1.5, the error rate for $k_n = 30$ with the test

statistic $TA2$ is around 2.8% and the detection rate is 84%. In contrast, for the test statistic $TA1$, the error rate is around 3% and the detection rate is 60%. It can also be seen that, for low values of k_n in $TA1$, there are low error rates but with low detection power. For example, for $q = 1.5$, in the case of $k_n = 20$, there is an error rate around 1.4% and a detection rate of 34%. In contrast for $TA2$ we observe 44% of detection.

Besides for the optimal critical value $q = 1.2322$, with $k_n = 30$ we have an error rate of 5% and a detection power of 90% for the test statistic $TA2$ and the test statistic $TA1$, we observe an error rate of 6% and a detection rate of 70%. Figure 4 shows the detection and error rates for a fixed critical value of 1.5. Figure 5 shows the detection and error rates of a simulated critical value $q = 1.2322$, that were obtained through the respective ROC curves in Figure 6.

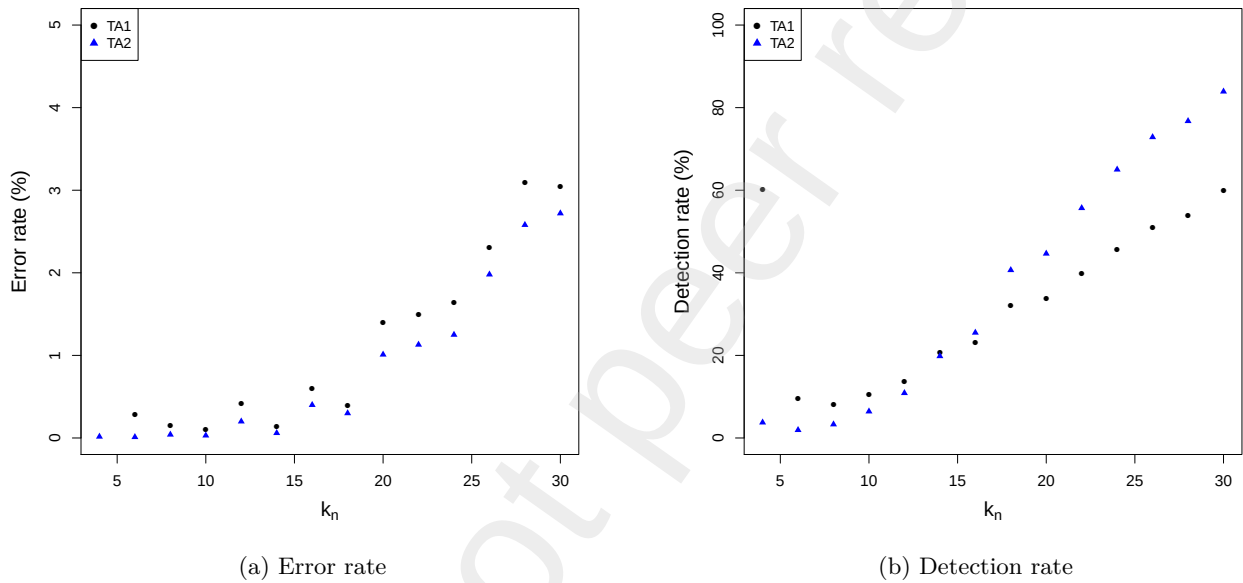
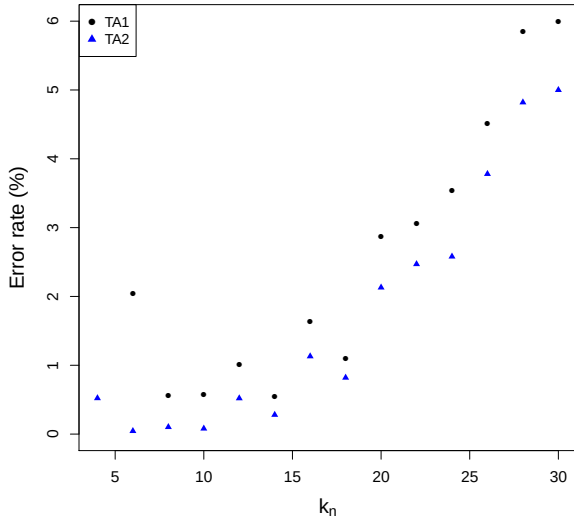


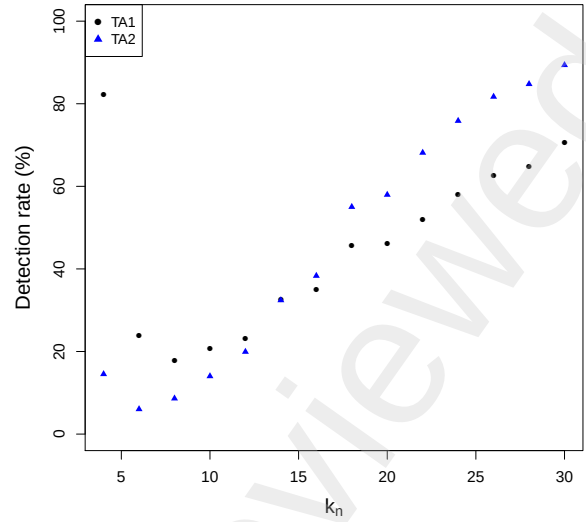
Figure 4 – Detection and error rates with critical value 1.5 and different values of k_n

We repeat our simulation for $n = 2000$ and the same pairs of k_n and m_n already mentioned above. The conclusion is similar to $n = 1000$, with respect to the average quadratic distance of the statistic, as we can see in Figure 7. Figure 8 shows the detection and error rates for a fixed critical value of 1.5. Table 2 shows the detection and error rates for two different critical values, which lead us to conclude that $TA2$ is better. For example, for $q = 1.3663$ we have an error rate of 5% and a detection power of 91% for the test statistic $TA2$. With the test statistic $TA1$, we observe an error rate of 5.8% and a detection rate of 66%. Figure 9 shows the detection and error rates for a fixed $q = 1.3663$, obtained by simulation, with the ROC curves in Figure 10. It can be seen that as k_n increases the error rate for $TA1$ is higher than for $TA2$ and, for the detection rate, as k_n increases, $TA2$ has a higher detection rate than $TA1$. Again $TA2$ seems to be better than $TA1$.

We can observe in Figure 11 that the quadratic distance of the statistics has a similar behavior

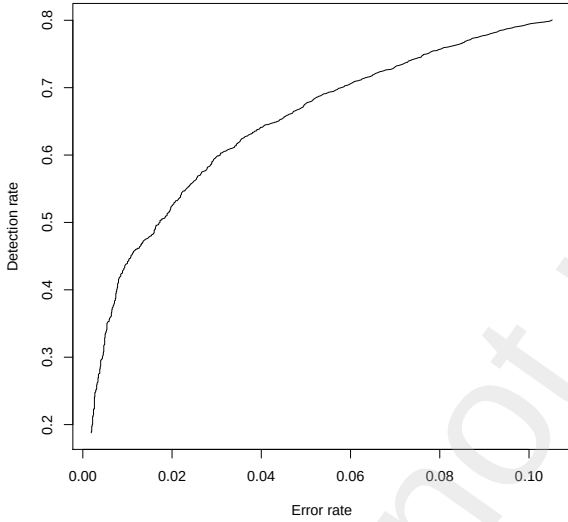


(a) Error rate

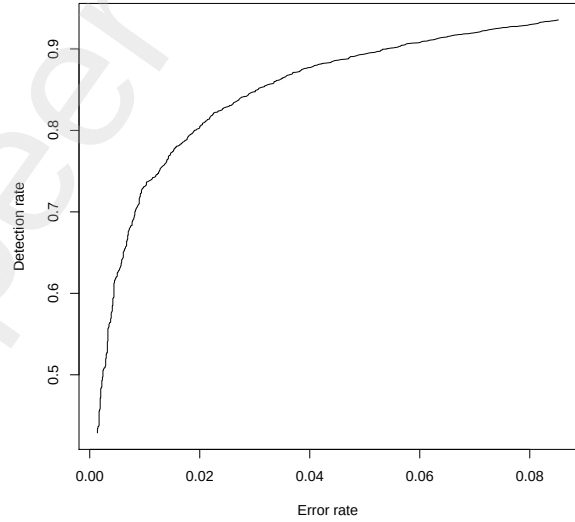


(b) Detection rate

Figure 5 – Detection and error rates with critical value 1.2322



(a) ROC curve TA1



(b) ROC curve TA2

Figure 6 – ROC curve for $TA1$ and $TA2$ statistics.

to the previous analysis. We are interested in the largest quadratic distance so for the pairs of k_n and m_n already mentioned above and $n = 5000$ we will do our study for the aforementioned reasons. Figure 12 shows the detection and error rates for a fixed critical value 1.5. Figure 13 shows the detection and error rate considering a simulated critical value obtained by ROC curve Figure 14.

Table 3 shows that for a fixed critical value 1.5, we see that, the error rate for $k_n = 70$ with the test statistic $TA2$ is around 3.9% and the detection rate of 90%. In contrast, for the test statistic $TA1$, the error rate is around 4.3% and detection rate of 56%. It can also be seen that, for low values of k_n in $TA1$, there are low error rates but with low detection power. For example, in the case of $k_n = 48$ with the test statistic $TA1$, there is an error rate around 1.2% and a detection rate of 32%.

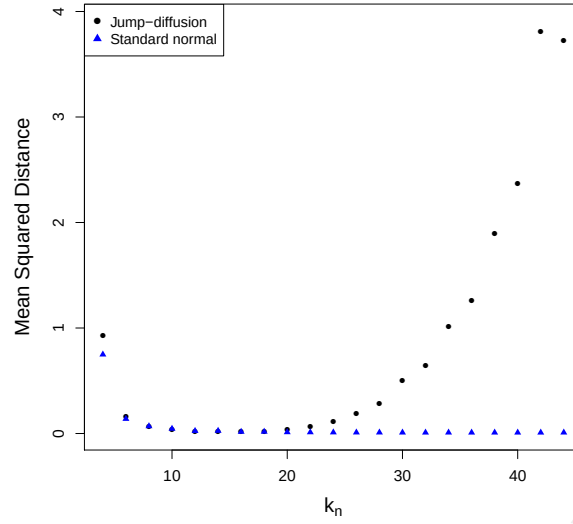


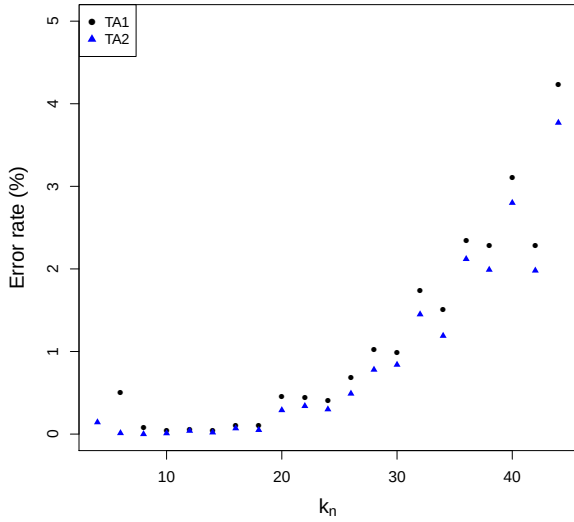
Figure 7 – Mean Squared Distance of the two test statistics for different values of k_n , in data generated from the two models with $n = 2000$.

Table 2 – Detection and error rates for k_n values between 4 and 44 for fixed critical value $q = 1.5$ and simulated the critical value $q = 1.3663$, in 10000 replicas of data generated with $n = 2000$.

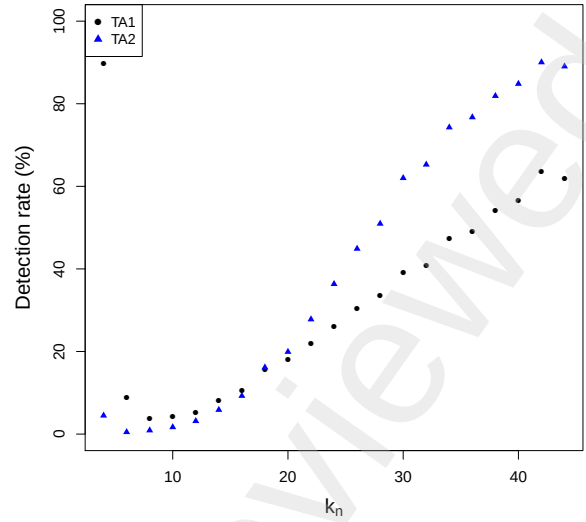
k_n	m_n	$q = 1.5$				$q = 1.3663$			
		Error rate		Detection rate		Error rate		Detection rate	
		$TA1$	$TA2$	$TA1$	$TA2$	$TA1$	$TA2$	$TA1$	$TA2$
4	2	0.7289	0.0014	0.8966	0.0449	0.8380	0.0090	0.9519	0.1101
6	3	0.0050	0.0001	0.0877	0.0049	0.0164	0.0002	0.1565	0.0110
8	4	0.0008	0.0000	0.0387	0.0090	0.0018	0.0000	0.0688	0.0162
10	5	0.0004	0.0001	0.0424	0.0168	0.0011	0.0003	0.0681	0.0288
12	5	0.0006	0.0004	0.0530	0.0316	0.0015	0.0008	0.0793	0.0512
14	8	0.0004	0.0002	0.0806	0.0587	0.0006	0.0003	0.1223	0.0892
16	7	0.0010	0.0007	0.1066	0.0926	0.0018	0.0010	0.1471	0.1277
18	11	0.0011	0.0005	0.1572	0.1615	0.0021	0.0008	0.2067	0.2145
20	9	0.0046	0.0029	0.1799	0.1992	0.0063	0.0041	0.2227	0.2533
22	11	0.0044	0.0034	0.2201	0.2779	0.0069	0.0051	0.2713	0.3444
24	12	0.0041	0.0030	0.2596	0.3635	0.0075	0.0056	0.3168	0.4325
26	13	0.0068	0.0049	0.3052	0.4489	0.0110	0.0076	0.3589	0.5166
28	13	0.0103	0.0078	0.3350	0.5095	0.0152	0.0116	0.3876	0.5730
30	16	0.0099	0.0084	0.3908	0.6201	0.0158	0.0125	0.4465	0.6761
32	15	0.0174	0.0145	0.4094	0.6527	0.0249	0.0208	0.4670	0.7058
34	18	0.0151	0.0119	0.4726	0.7428	0.0223	0.0183	0.5326	0.7878
36	17	0.0234	0.0212	0.4911	0.7673	0.0341	0.0295	0.5451	0.8075
38	20	0.0228	0.0199	0.5409	0.8189	0.0310	0.0263	0.5943	0.8515
40	20	0.0311	0.0280	0.5648	0.8483	0.0430	0.0381	0.6188	0.8776
42	26	0.0228	0.0198	0.6370	0.9004	0.0327	0.0274	0.6876	0.9205
44	21	0.0423	0.0377	0.6201	0.8905	0.0576	0.0500	0.6659	0.9118

The critical value obtained by ROC curve is $q = 1.3663$. With this, we have an error rate of 4.9% and a detection power of 91% for the test statistic $TA2$. With the test statistic using the $TA1$, we observe an error rate of 5.5% and a detection rate of 59%.

We will use the $TA2$ test statistic, which uses the truncated variance estimator defined in (13), since it presented better results in relation to low error rate and high detection power. The ideal is to

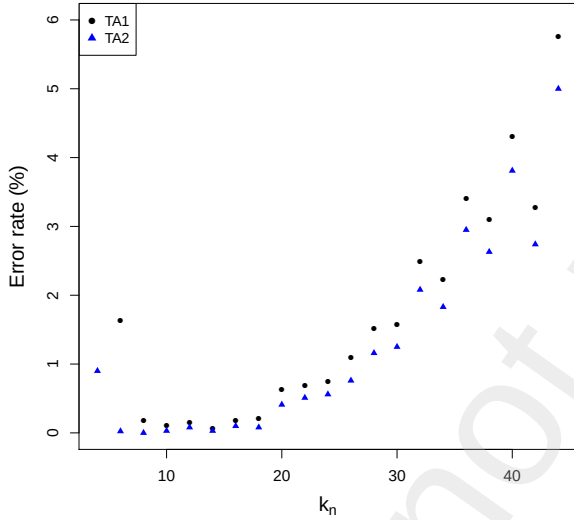


(a) Error rate

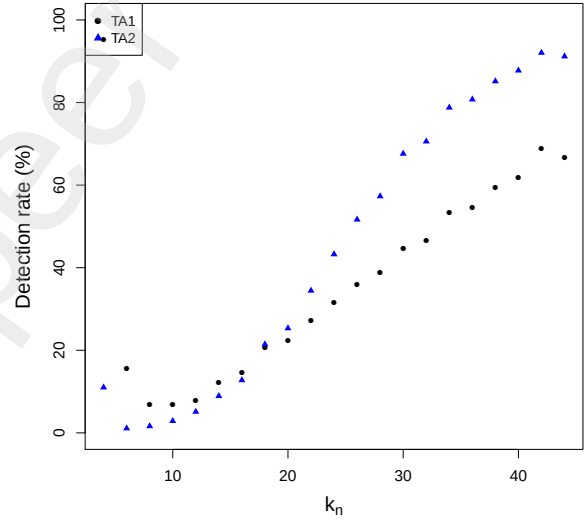


(b) Detection rate

Figure 8 – Detection and error rates with critical value 1.5



(a) Error rate



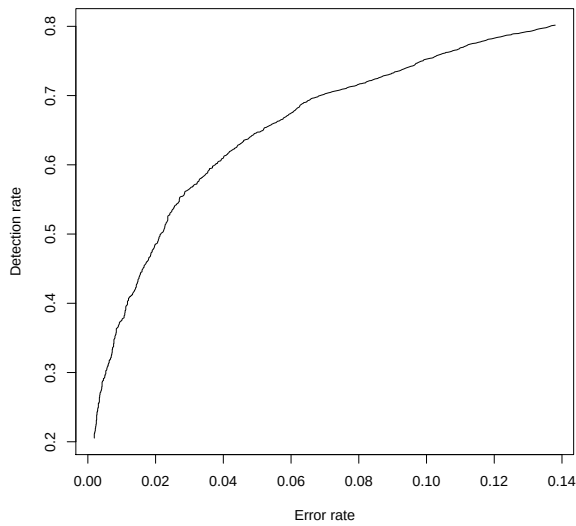
(b) Detection rate

Figure 9 – Detection and error rates with critical value 1.3663

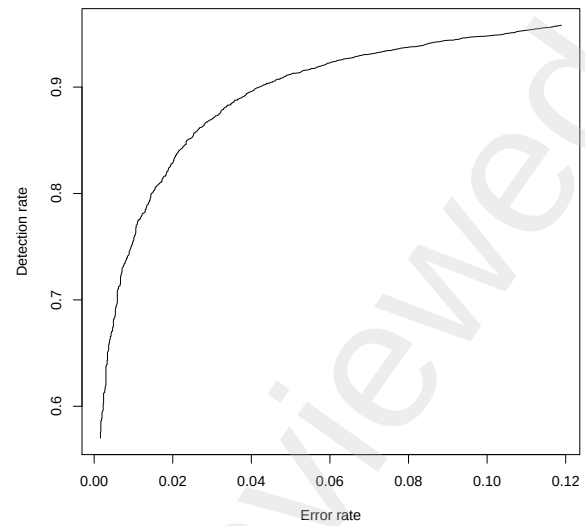
choose a k_n that shows a low error rate and high detection power. In contrast, for the test statistic $TA1$, there is high error rate and low detection power for most k_n values. Table 4 shows a summary of the results for the 3 possible sample sizes. Note that in Tables 1,2 and 3 for values $k_n = 4$ and $k_n = 6$ the error and detection rate shows inconsistent values, this is because it has a high false alarm rate and not so much detection power. Therefore, we do not recommend using low values of k_n .

5 Practical Considerations

The conditions in Theorem 1 and as discussed in Kong (2017), for a finite sample n , k_n should be smaller than $\sqrt{n}$ and m_n should be smaller than k_n , with k_n/m_n ranging from 1.5 to 2.5 and having



(a) ROC curve TA1



(b) ROC curve TA2

Figure 10 – ROC curve for $TA1$ and $TA2$ statistic.

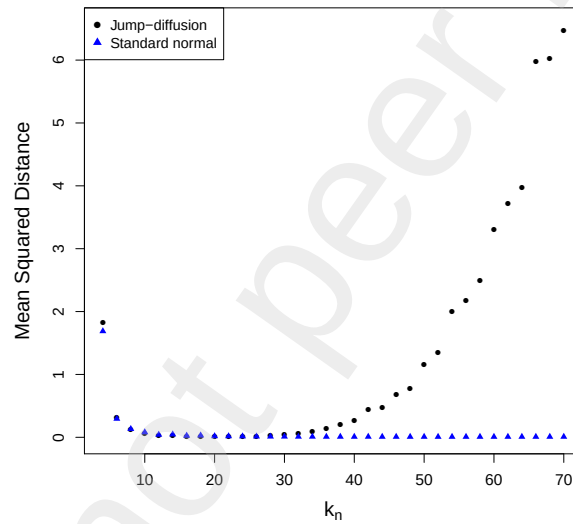
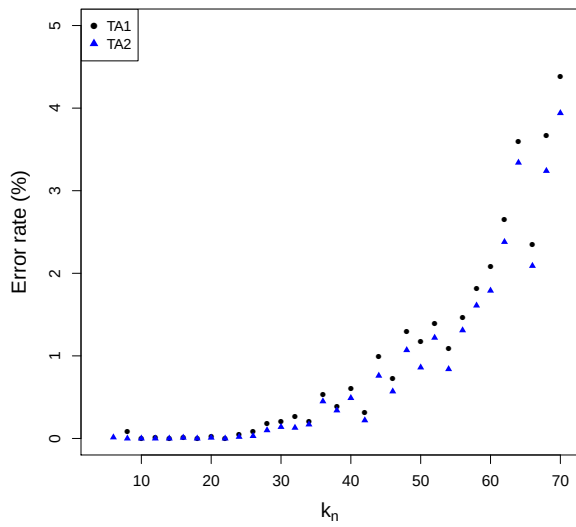
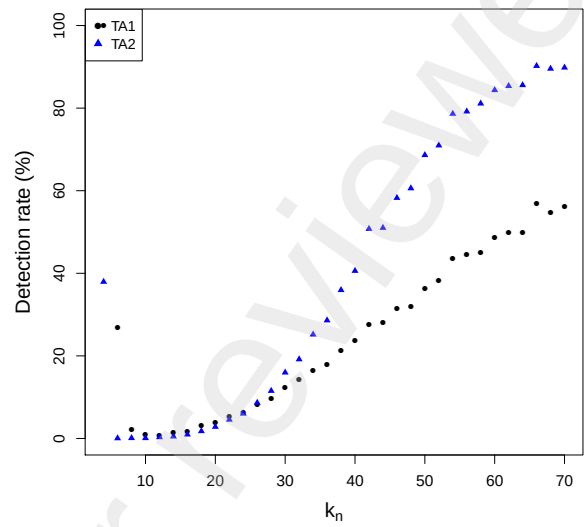


Figure 11 – Mean Squared Distance of the two test statistics for different values of k_n , in data generated from the two models with $n = 5000$.

and increasing trend. We will use the $TA2$ test statistic, which uses the truncated variation estimator defined in (13), since it presented better results in relation to low error rate and high detection power. The ideal is to choose a k_n that shows a low error rate and high detection power. In contrast, for the test statistic $TA1$, there is high error rate and low detection power for most k_n values.

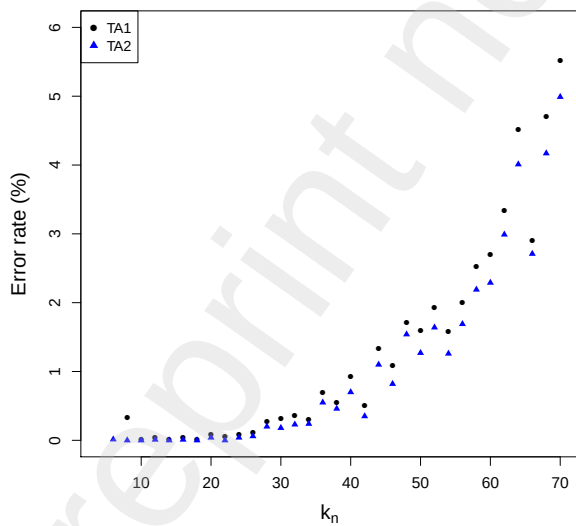


(a) Error rate

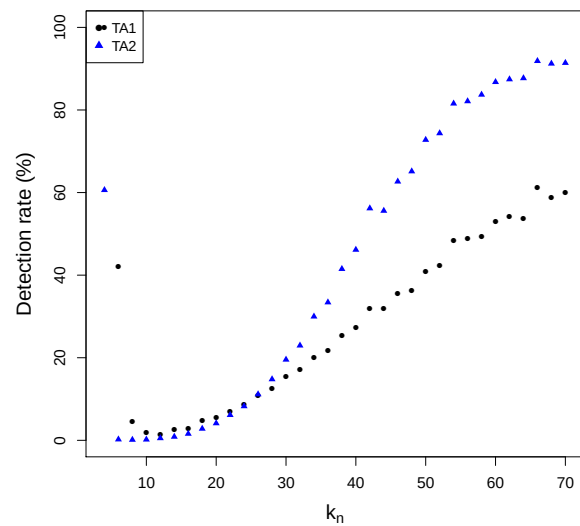


(b) Detection rate

Figure 12 – Detection and error rates with critical value 1.5



(a) Error rate



(b) Detection rate

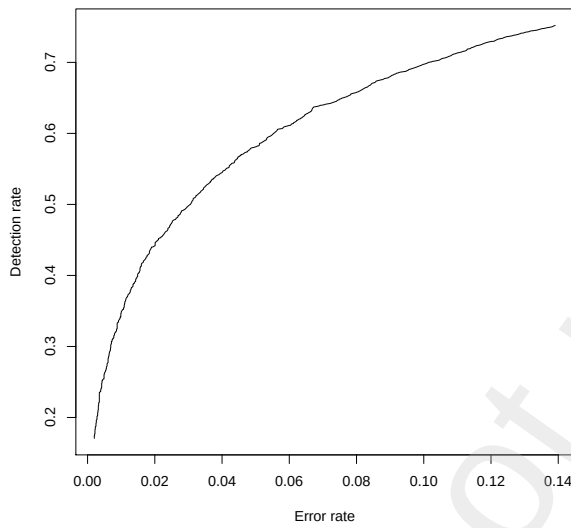
Figure 13 – Detection and error rates with critical value 1.3943

Table 3 – Detection and error rates for k_n values between 4 and 70 for fixed critical value $q = 1.5$ and simulated the critical value $q = 1.3943$, in 10000 replicas of data generated with $n = 5000$.

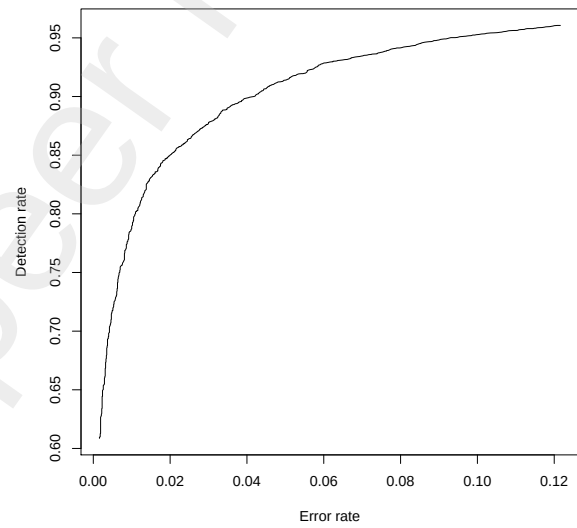
k_n	m_n	$q = 1.5$				$q = 1.3943$			
		Error rate		Detection rate		Error rate		Detection rate	
		$TA1$	$TA2$	$TA1$	$TA2$	$TA1$	$TA2$	$TA1$	$TA2$
4	2	1.0000	0.1851	1.0000	0.3794	1.0000	0.4053	1.0000	0.6061
6	3	0.1101	0.0001	0.2681	0.0005	0.2176	0.0001	0.4208	0.0022
8	4	0.0009	0.0000	0.0226	0.0010	0.0034	0.0000	0.0452	0.0013
10	5	0.0000	0.0000	0.0094	0.0009	0.0001	0.0000	0.0179	0.0019
12	5	0.0001	0.0000	0.0082	0.0031	0.0004	0.0001	0.0145	0.0052
14	8	0.0000	0.0000	0.0150	0.0048	0.0001	0.0000	0.0261	0.0087
16	7	0.0001	0.0001	0.0171	0.0096	0.0004	0.0001	0.0279	0.0159
18	11	0.0000	0.0000	0.0314	0.0177	0.0001	0.0000	0.0481	0.0281
20	9	0.0002	0.0001	0.0387	0.0282	0.0008	0.0004	0.0548	0.0413
22	11	0.0000	0.0000	0.0536	0.0455	0.0006	0.0000	0.0699	0.0613
24	12	0.0005	0.0002	0.0638	0.0602	0.0009	0.0004	0.0875	0.0826
26	13	0.0008	0.0003	0.0815	0.0864	0.0012	0.0006	0.1080	0.1115
28	13	0.0018	0.0010	0.0978	0.1153	0.0028	0.0020	0.1246	0.1479
30	16	0.0020	0.0014	0.1233	0.1597	0.0031	0.0018	0.1538	0.1954
32	15	0.0027	0.0013	0.1432	0.1917	0.0036	0.0023	0.1705	0.2296
34	18	0.0021	0.0017	0.1649	0.2518	0.0030	0.0024	0.2011	0.2998
36	17	0.0053	0.0045	0.1800	0.2860	0.0070	0.0055	0.2174	0.3341
38	20	0.0039	0.0034	0.2135	0.3594	0.0055	0.0046	0.2530	0.4150
40	20	0.0061	0.0049	0.2368	0.4059	0.0093	0.0070	0.2742	0.4615
42	26	0.0031	0.0022	0.2755	0.5076	0.0050	0.0035	0.3185	0.5618
44	21	0.0099	0.0076	0.2806	0.5101	0.0133	0.0110	0.3189	0.5559
46	25	0.0072	0.0057	0.3156	0.5826	0.0108	0.0082	0.3556	0.6267
48	23	0.0129	0.0107	0.3206	0.6058	0.0171	0.0154	0.3626	0.6514
50	28	0.0117	0.0086	0.3629	0.6863	0.0160	0.0127	0.4086	0.7277
52	27	0.0139	0.0122	0.3823	0.7097	0.0193	0.0164	0.4226	0.7437
54	34	0.0109	0.0084	0.4359	0.7863	0.0158	0.0126	0.4834	0.8156
56	31	0.0147	0.0131	0.4448	0.7919	0.0200	0.0169	0.4888	0.8212
58	30	0.0182	0.0161	0.4508	0.8111	0.0252	0.0219	0.4944	0.8371
60	34	0.0208	0.0179	0.4876	0.8437	0.0270	0.0229	0.5301	0.8677
62	33	0.0265	0.0238	0.4999	0.8536	0.0334	0.0299	0.5411	0.8743
64	30	0.0360	0.0334	0.4983	0.8557	0.0451	0.0401	0.5380	0.8770
66	40	0.0235	0.0209	0.5697	0.9022	0.0291	0.0271	0.6127	0.9186
68	35	0.0367	0.0324	0.5480	0.8955	0.0470	0.0417	0.5882	0.9123
70	33	0.0438	0.0394	0.5613	0.8982	0.0552	0.0499	0.5994	0.9140

Table 4 – Error and detection rates of $TA2$ statistic for a critical value fixed $q = 1.5$ and optimal critical values for some scenarios.

n	k_n	m_n	$q = 1.5$		Optimal q	
			Error rate	Detection rate	Error rate	Detection rate
1000	30	16	0.0272	0.8390	0.05 ($q = 1.2322$)	0.8937
2000	44	21	0.0377	0.8905	0.05 ($q = 1.3663$)	0.9118
5000	70	33	0.0394	0.8982	0.049 ($q = 1.3943$)	0.9140



(a) ROC curve TA1



(b) ROC curve TA2

Figure 14 – ROC curve for $TA1$ and $TA2$ statistic.

6 Quantile Analysis

The purpose is to find the theoretical quantile of our test statistic as stated above; for this we must find the cumulative distribution function of our statistics Cramér-von Mises given by:

$$\begin{aligned} P(T_A^n \leq x) &= P\left(d^2 \sqrt{[n/2k_n] m_n} \leq x\right) \\ &= P\left(\sqrt{[n/2k_n] m_n} \int_{-\infty}^{\infty} [\hat{F}'_n(\tau) - \Phi(\tau)]^2 d\Phi(\tau) \leq x\right), \tau \in R, x \geq 0. \end{aligned} \quad (15)$$

where $\hat{F}'_n(\tau)$ is the empirical c.d.f. given in (14) and $\Phi(\tau)$ is the c.d.f of standard normal. Here, $\Phi'(\tau) = d\Phi(\tau) = \phi(\tau)d\tau$. The expression (15) do not have a closed form (for details see the Appendix). So we have to resort to numerical methods to calculate it.

Our simulation study shows that the c.d.f. of our $TA2$ test statistic is approximately a gamma distribution with parameters of shape and scale, a and b , respectively. The gamma distribution has as density

$$f(x) = \frac{1}{b^a \Gamma(a)} x^{a-1} \exp^{-x/b}, \quad x \geq 0, \quad a > 0, \text{ and } b > 0.$$

The mean and variance are $E[x] = ab$ and $\text{Var}[x] = ab^2$. Since $E[x] = ab$ then $a = \frac{E[x]}{b}$. On the other hand $\text{Var}[x] = ab^2 = \frac{E[x]}{b} b^2 = E[x]b \Rightarrow b = \text{Var}[x] \frac{1}{E[x]}$. Thus, the gamma c.d.f. is given by

$$\begin{aligned} F(x) &= \int_0^x f(t)dt = \int_0^x \frac{1}{b^a \Gamma(a)} t^{a-1} e^{-t/b} dt = \frac{1}{\Gamma(a)} \int_0^x \frac{t^{a-1}}{b^a} e^{-t/b} dt \\ &= \frac{1}{\Gamma(a)} \int_0^x \left(\frac{t}{b}\right)^{a-1} e^{-t/b} dt. \end{aligned}$$

Making the transformation $y = t/b \Rightarrow dy = dt/b$, we have

$$F(x) = \frac{1}{\Gamma(a)} \int_0^x y^{a-1} e^{-y} dy.$$

The gamma c.d.f. involves the incomplete gamma function given by

$$\int_0^x y^{a-1} e^{-y} dy = \gamma(a, x).$$

So finally,

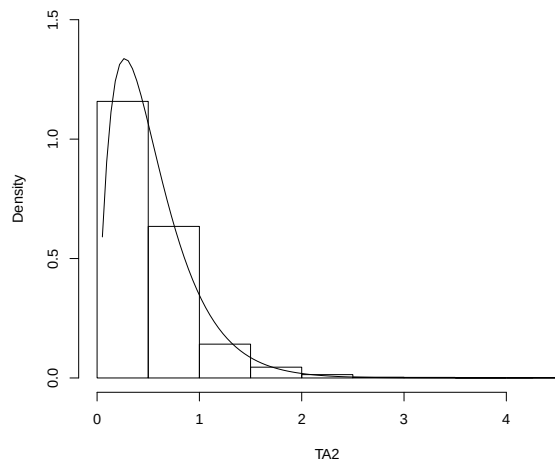
$$F(x) = \frac{\gamma(a, x)}{\Gamma(a)}, \quad a > 0.$$

Then we can write, $TA2 \sim^a \text{Gamma}(a, b)$. For large sample sizes, we observe that $a \approx 2$ and $b \approx 0.25$. This is

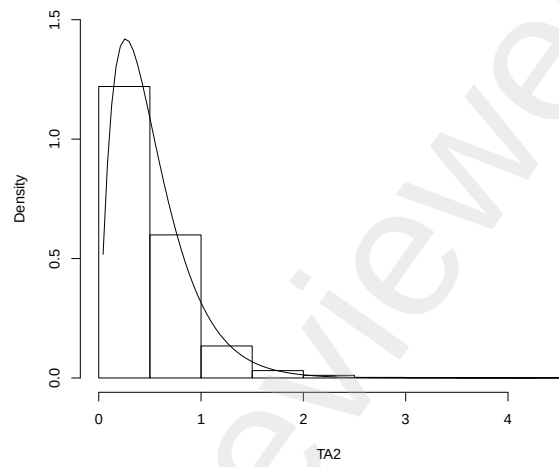
$$TA2 \sim^a \text{Gamma}(2, 0.25),$$

with $E[TA2] = 0.5$ and $\text{Var}[TA2] = 0.125$.

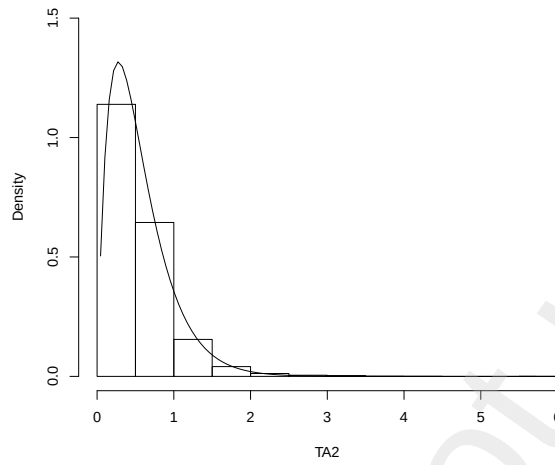
In Figure 19 (Appendix B) we observe that the sampling c.d.f. of $TA2$ and Gamma c.d.f. are very close. In Figure 15 we have the histograms for the sample values and the approximation with the true density, noticing the good agreement.



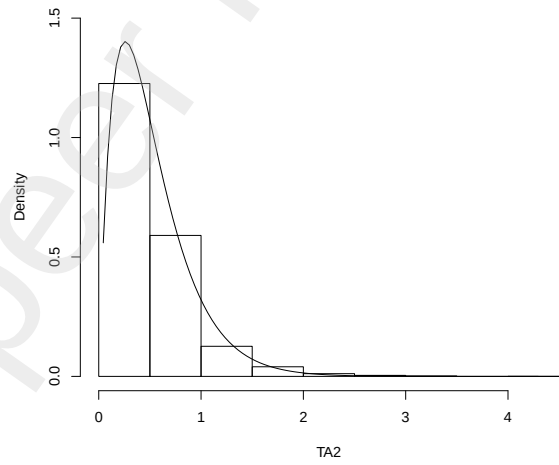
(a) $k_n = 40, n = 2000$.



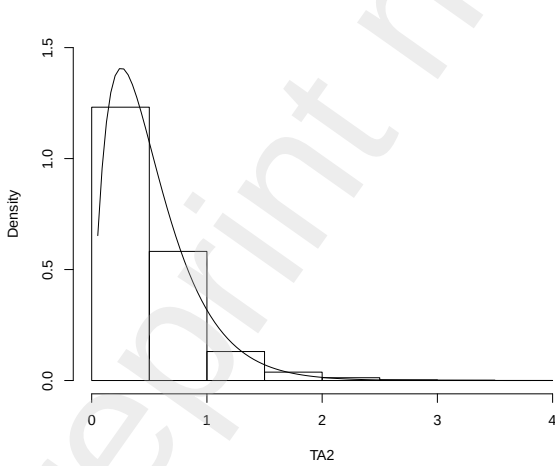
(b) $k_n = 42, n = 2000$.



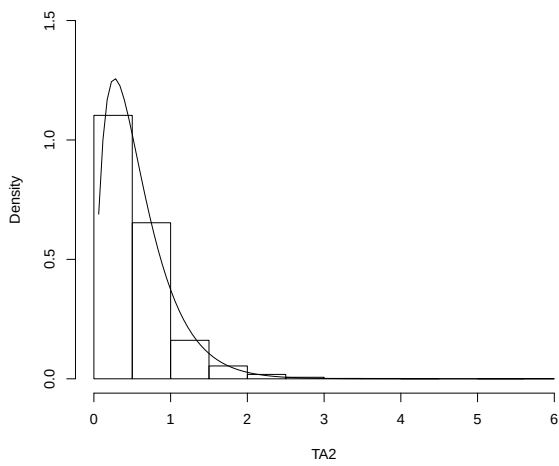
(c) $k_n = 44, n = 2000$.



(d) $k_n = 66, n = 5000$.



(e) $k_n = 68, n = 5000$.



(f) $k_n = 70, n = 5000$.

Figure 15 – Histograms of the sample density function for different sizes of k_n and n , compared with the gamma density.

Table 5 – Comparison between sample quantile and gamma quantile.

n	k_n	m_n	a	b	$q - \text{gamma}(a, b)$ 0.95	Sample Quantile 0.95
500	18	11	2.254	0.190	0.982	0.970
	20	11	2.109	0.238	1.174	1.165
	22	12	1.938	0.274	1.273	1.243
1000	26	13	2.083	0.231	1.128	1.116
	28	13	1.993	0.267	1.266	1.270
	30	19	2.064	0.228	1.106	1.084
2000	40	18	1.922	0.283	1.308	1.296
	42	22	1.972	0.267	1.259	1.246
	44	22	1.953	0.282	1.320	1.304
5000	66	36	1.993	0.261	1.239	1.251
	68	38	1.974	0.264	1.241	1.247
	70	34	1.885	0.308	1.403	1.380
10000	96	62	1.947	0.253	1.181	1.179
	98	48	1.894	0.301	1.379	1.357
	100	54	1.972	0.284	1.334	1.322

7 Simulation study

In this section, we conduct simulations studies to check the performance of the test. We consider the following two models: the first is given by,

$$\begin{aligned} X_t &= X_0 + \int_0^t \sqrt{c_s} dW_s + 0.5Z_t, \quad 0 \leq t \leq T, \\ c_t &= c_0 + \int_0^t 0.03(1.0 - c_s) ds + 0.15 \int_0^t \sqrt{c_s} dW'_s \end{aligned}$$

where Z_t is a skewed β -stable Lévy process. The volatility c_t is a square root diffusion process which is widely used in financial applications. The parameter in c_t is specified as in Jacob and Todorov (2014). The second model is

$$\tilde{X}_t = X_{t-1} + a_t, \quad a_t \sim N(0, 1), \quad (16)$$

where X_0 is the initial value that should be defined.

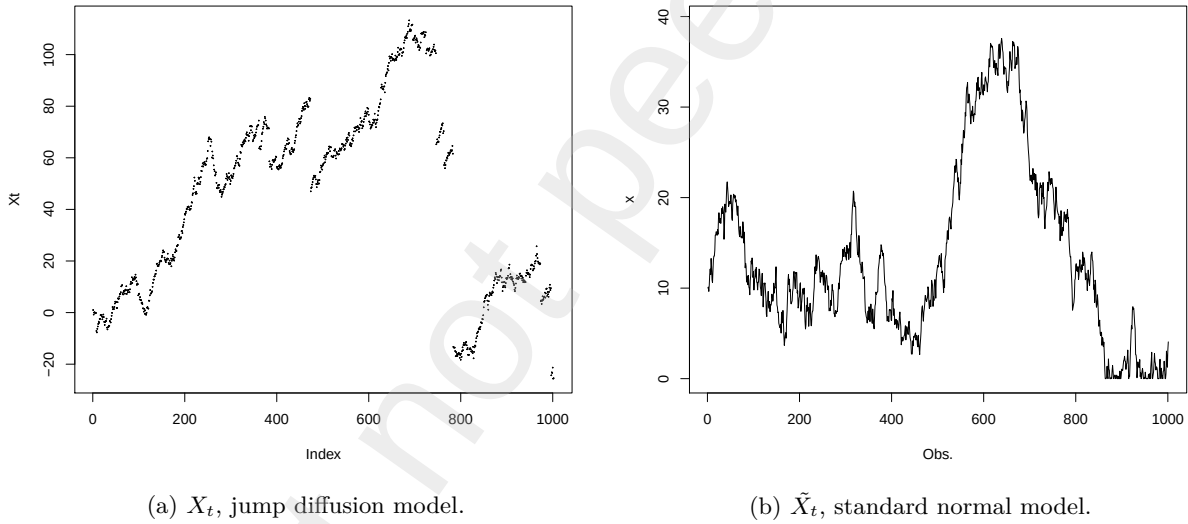


Figure 16 – Example of time series generate with the two models.

We considered 10000 replicas for sample sizes $n = 500, 1000, 2000, 5000$ and 10000. For the truncation of the increments, as is typical in the literature, we set $\alpha = 3.0$ and $\bar{w} = 0.49$. Hence for the sampling frequencies mentioned above, we set the pair of (k_n, m_n) to be $(22, 12)$, $(30, 19)$, $(44, 22)$, $(70, 34)$ and $(100, 54)$ with k_n/m_n ranging from 1.5 to 2.15 and having an increasing trend.

In Table 6 we can observe that the test power increases as the sample size increases, and also for a fixed n the test power is bigger for largest k_n values. The test power was calculated with the gamma quantile values, and we note that these values are very close to the sample quantile.

To evaluate the performance of our test, we compare with the Kolmogorov-Smirnov (KS) test, that also measures discrepancy between distributions. In Table 7 we note that our test shows better

Table 6 – Comparison between the sampling and theoretical quantiles for $TA2$.

n	k_n	m_n	a	b	$q - \text{gamma}(a, b)$ 0.95	Sample Quantile 0.95	Test Power
500	4	2	6.323	0.087	0.962	0.905	0.564
	6	3	2.302	0.165	0.865	0.689	0.615
	8	4	3.592	0.096	0.693	0.684	0.818
	18	11	2.254	0.190	0.982	0.968	0.956
	20	11	2.109	0.238	1.174	1.165	0.928
	22	12	1.938	0.274	1.273	1.243	0.930
1000	4	2	13.991	0.049	1.017	1.004	0.875
	6	3	6.924	0.060	0.709	0.680	0.958
	8	4	5.350	0.064	0.623	0.621	0.981
	26	13	2.083	0.231	1.128	1.116	0.999
	28	13	1.993	0.267	1.266	1.270	0.997
	30	19	2.064	0.228	1.106	1.084	0.999
2000	4	2	27.702	0.032	1.202	1.196	0.990
	6	3	14.850	0.033	0.736	0.731	0.999
	8	4	9.147	0.041	0.612	0.607	0.999
	40	18	1.922	0.283	1.308	1.296	1
	42	22	1.972	0.267	1.259	1.246	1
	44	22	1.953	0.282	1.320	1.304	1
5000	4	2	69.609	0.019	1.633	1.631	1
	6	3	35.490	0.020	0.921	0.920	1
	8	4	21.383	0.023	0.694	0.694	1
	66	36	1.993	0.261	1.239	1.251	1
	68	38	1.974	0.264	1.241	1.247	1
	70	34	1.885	0.308	1.403	1.380	1
10000	4	2	132.233	0.014	2.155	2.154	1
	6	3	70.155	0.013	1.171	1.170	1
	8	4	41.810	0.015	0.839	0.838	1
	96	62	1.947	0.253	1.181	1.179	1
	98	48	1.894	0.301	1.379	1.357	1
	100	54	1.972	0.284	1.334	1.322	1

results. Besides, the KS test power decreases and the error rate increases as n increases. For $n = 500$ the KS test power is bigger than $TA2$ power, however $TA2$ has error rate 5% and the power of the KS test has an error rate bigger than 76%. For larger sample sizes, $TA2$ shows better power with low error rate. In Table 8 we can find the quantile values of gamma and sample distribution for different probabilities. Note that the theoretical quantile compared with the sample quantile are closer for probabilities higher than 0.90. Frequently, in hypotheses testing problems, it is used a significance level less than 0.10, so we can use the gamma distribution to choose a critical value for the test.

Table 7 – Comparison between our test statistic $TA2$ and KS test.

n	k_n	m_n	a	b	$q - \text{gamma}(a, b)$ 0.95	Sample Quantile 0.95	Power $TA2$	Test Power KS	Error Rate KS
500	4	2	6.323	0.087	0.962	0.905	0.564	0.958	0.763
	6	3	2.302	0.165	0.865	0.689	0.615	0.958	0.963
	8	4	3.592	0.096	0.693	0.684	0.818	0.959	0.987
	18	11	2.254	0.190	0.982	0.968	0.956	0.960	0.999
	20	11	2.109	0.238	1.174	1.165	0.928	0.958	1
	22	12	1.938	0.274	1.273	1.243	0.930	0.960	1
1000	4	2	13.991	0.049	1.017	1.004	0.875	0.911	0.691
	6	3	6.924	0.060	0.709	0.680	0.958	0.912	0.913
	8	4	5.350	0.064	0.623	0.621	0.981	0.911	0.974
	26	13	2.083	0.231	1.128	1.116	0.999	0.913	1
	28	13	1.993	0.267	1.266	1.270	0.997	0.914	1
	30	19	2.064	0.228	1.106	1.084	0.999	0.913	1
2000	4	2	27.702	0.032	1.202	1.196	0.990	0.836	0.528
	6	3	14.850	0.033	0.736	0.731	0.999	0.836	0.897
	8	4	9.147	0.041	0.612	0.607	0.999	0.836	0.977
	40	18	1.922	0.283	1.308	1.296	1	0.836	1
	42	22	1.972	0.267	1.259	1.246	1	0.838	1
	44	22	1.953	0.282	1.320	1.304	1	0.837	1
5000	4	2	69.609	0.019	1.633	1.631	1	0.635	0.462
	6	3	35.490	0.020	0.921	0.920	1	0.635	0.818
	8	4	21.383	0.023	0.694	0.694	1	0.635	0.959
	66	36	1.993	0.261	1.239	1.251	1	0.638	1
	68	38	1.974	0.264	1.241	1.247	1	0.637	1
	70	34	1.885	0.308	1.403	1.380	1	0.637	1

Table 8 – Percentage points for $TA2$. Entries in the table are x such that $P\{d^2\sqrt{[n/2k_n]m_n} \leq x\} = P\{TA2 \leq x\} = p$. The upper number in a double entry is the critical value calculated by using the gamma distribution; the lower is that based on the sample distribution

n	Percentage points for the following values of p :																n
	0.01	0.025	0.05	0.1	0.15	0.20	0.25	0.50	0.75	0.80	0.85	0.90	0.95	0.975	0.99	0.999	
500	0.037	0.061	0.090	0.137	0.177	0.215	0.251	0.443	0.717	0.798	0.901	1.041	1.273	1.498	1.788	2.495	500
	0.121	0.145	0.167	0.201	0.231	0.259	0.285	0.430	0.650	0.722	0.825	0.964	1.243	1.589	2.054	3.194	
1000	0.037	0.059	0.087	0.129	0.164	0.198	0.230	0.397	0.633	0.702	0.790	0.909	1.106	1.297	1.542	2.138	1000
	0.107	0.129	0.152	0.186	0.212	0.235	0.259	0.382	0.569	0.641	0.727	0.859	1.084	1.361	1.754	2.700	
2000	0.040	0.064	0.095	0.143	0.185	0.224	0.262	0.461	0.745	0.829	0.935	1.080	1.320	1.553	1.853	2.583	2000
	0.117	0.142	0.169	0.206	0.238	0.267	0.296	0.443	0.679	0.759	0.863	1.023	1.304	1.625	2.072	3.278	
5000	0.037	0.0624	0.096	0.145	0.189	0.230	0.270	0.482	0.785	0.875	0.989	1.145	1.403	1.654	1.978	2.769	5000
	0.119	0.148	0.176	0.217	0.250	0.279	0.306	0.463	0.712	0.794	0.907	1.090	1.380	1.693	2.235	3.403	
10000	0.040	0.066	0.097	0.147	0.189	0.228	0.267	0.468	0.754	0.840	0.947	1.093	1.334	1.569	1.871	2.606	10000
	0.118	0.141	0.169	0.209	0.242	0.270	0.298	0.449	0.692	0.770	0.880	1.045	1.322	1.610	2.048	3.252	

8 Real Data Analysis

In this section, we collect intraday transaction prices of the Google, Apple and Goldman Sachs (GS) stocks, respectively, from November 11th to November 12th in 2014, with a sampling frequency up to every 15 seconds. The transaction records that are outside the ranges of quotes, in our case 9:30 a.m to 4:00 p.m are excluded. There are in total 3122, 2898 and 2819 stock of prices, respectively. We aim to plot the observed test statistics against different values of k_n and m_n . Figure 17 shows the three

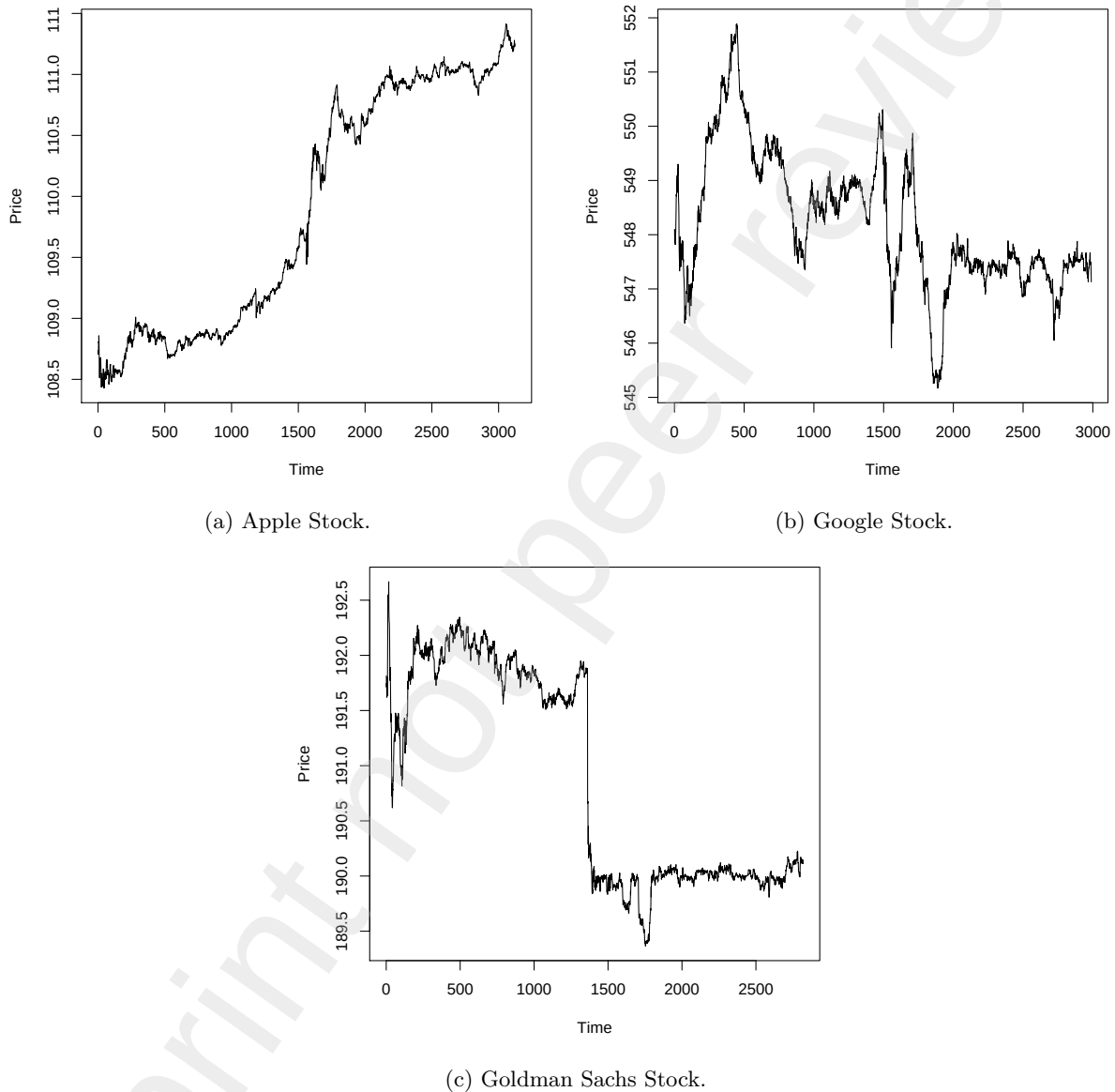


Figure 17 – Time series of the data from 9 : 30 a.m to 4 : 00 p.m.

different stock market data. Figure 18 shows the values of our test statistic, compared to the values of k_n . Additionally, we illustrate the quantile values at the significance level of 5% and 10%. The graphics on the left of Figure 18 consider the values of k_n between $[4, 55]$, $[4, 54]$ and $[4, 53]$ respectively. The graphics on the right show k_n values starting from 44, that is, we show an enlarged version of what this graphics controller looks like on the side. It can be observed that at a level of significance of 5%,

Apple stock rejects null hypotheses 88%, unlike Google and GS stocks that do not reject with 2% and 16% respectively. On the other hand, when we increase the level of significance to 10%, the percentage increases for all stocks, with Apple still rejecting with 92%, Google does not reject with 10% and GS also does not reject with 20%. This using all k_n values. For sufficiently large values of k_n (order of

Table 9 – Rejection rate of the test, considering all possible k_n values.

Stock	n	k_n	m_n	$\alpha = 5\%(q = 1.361)$	$\alpha = 10\%(q = 1.112)$
Apple	3122	[4,55]	[2,27]	88%	92%
Google	2989	[4,54]	[2,27]	2%	10%
GS	2819	[4,53]	[2,26]	16%	20%

magnitude considered in our study), our test statistic tends to detect more precisely the dynamics of the Apple and GS series as can be seen in Table 10. On the other hand, our test statistic even for large k_n values, it was not able to detect the dynamics of the Google series in the analyzed period, this may be due to the fact that the series presents an uncontrolled variability, see Figure 17 (b).

Table 10 – Rejection rate of the test, considering k_n values that are at least 44.

Stock	n	k_n	m_n	$\alpha = 5\%(q = 1.361)$	$\alpha = 10\%(q = 1.112)$
Apple	3122	[44,55]	[22,27]	100%	100%
Google	2989	[44,54]	[22,27]	9%	27%
GS	2819	[44,53]	[22,26]	70%	80%

9 Conclusion

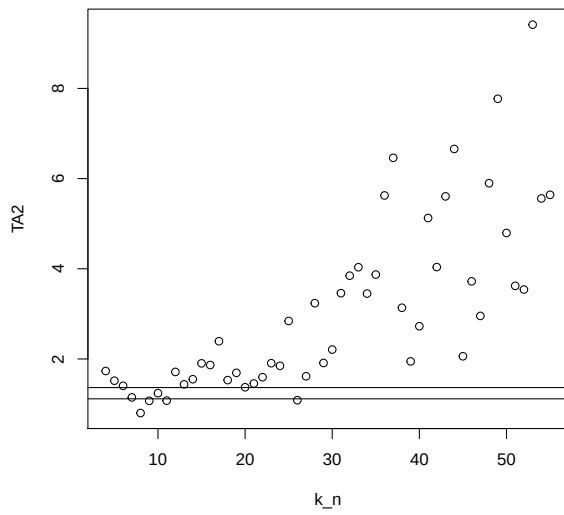
We construct a test statistic that capture the dynamics of a series using high frequency transaction prices. We compare the test power of our statistic with the KS test statistic and the performance of $TA2$ show better results. The simulation show that we can approximate our c.d.f. of the test statistic $TA2$ with the gamma distribution, so we can use it in practice to apply the test for financial strategies.

The empirical data analysis shows that our statistic $TA2$ is useful to identify that, increments in time series like Apple and GS stocks, follow a jumps-diffusion model for high frequency data of prices for the largest k_n values.

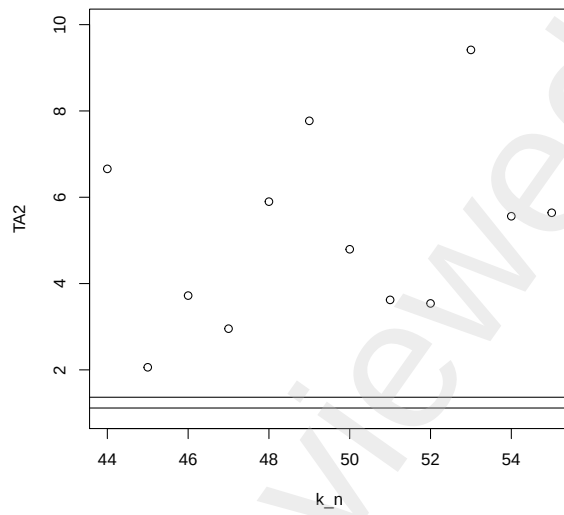
I believe that, nowadays, in finance and econometrics it is still an open problem, being able to anticipate and/or measure (Threshold) the size of the big losses that will occur on some day of trading on the stock exchange, or of big gains if that is the case, given the need to mitigate the risk of any portfolio in question.

Acknowledgements

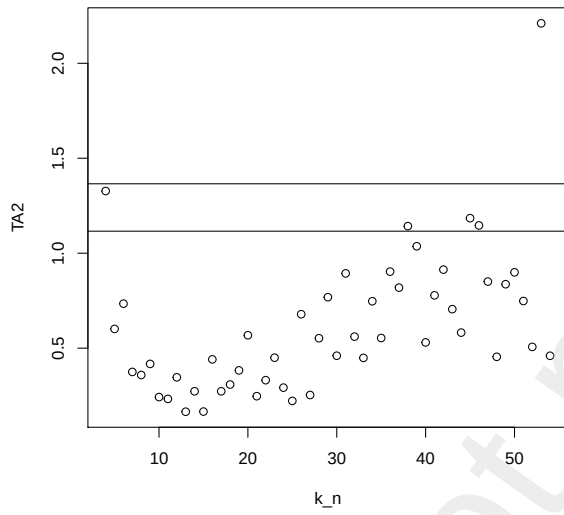
This work was financed by a fellowship supported by the International Cooperation Program CAPES/PDSE of Brazil, at the University of Chicago.



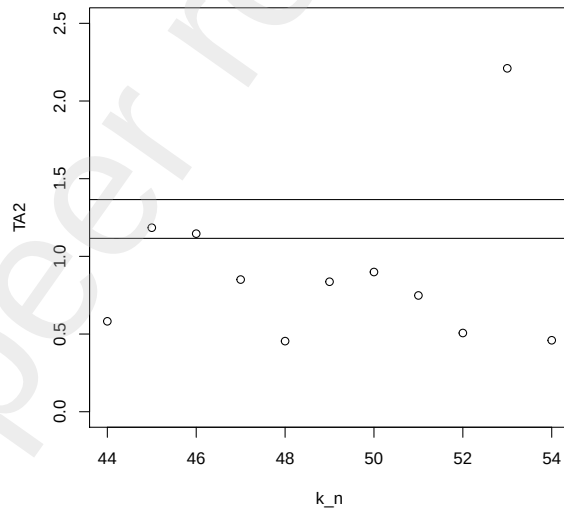
(a) Apple Stock, $k_n = 4, \dots, 55$.



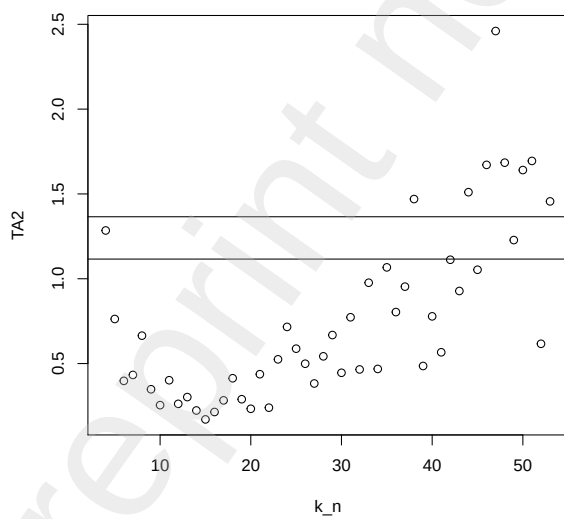
(b) Apple Stock, $k_n = 44, \dots, 55$.



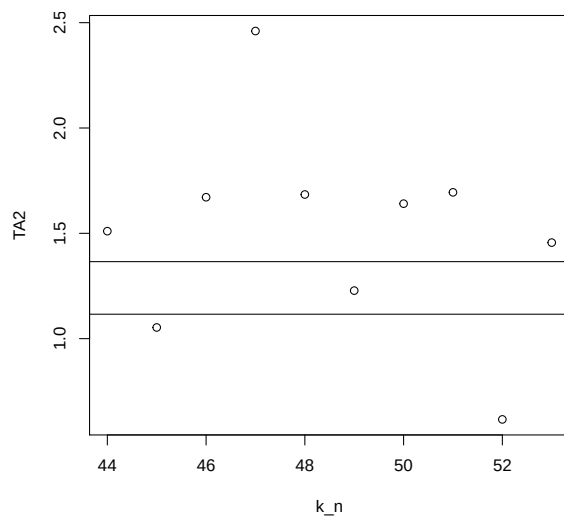
(c) Google Stock, $k_n = 4, \dots, 54$.



(d) Google Stock, $k_n = 44, \dots, 54$.



(e) Goldman Sachs Stock, $k_n = 4, \dots, 53$.



(f) Goldman Sachs Stock, $k_n = 44, \dots, 53$.

Figure 18 – $TA2$ statistic for different k_n .

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10 Appendix A

Cumulative distribution function of $TA2$. In order to make inferences and extract the maximum amount of information about the behavior of the dynamics of a time series or any data set; Identifying the density function and where the moments are from the cumulative distribution function has been one of the most important tasks in statistics to date. In this case; in order to identify the distribution of our test statistic $TA2$ analytically; that is; identify the core of the distribution resulting from the calculation of $TA2$; we can directly approximate it by some known distribution if we can identify its core, so we won't have to resort to computational power to calculate it. Let's see if this is true. We have,

$$P(d^2 \leq x) = P\left(\int_{-\infty}^{\infty} [\hat{F}'_n(\tau) - \Phi(\tau)]^2 d\Phi(\tau) \leq x\right), \quad \tau \in R, \quad x \geq 0.$$

Here, $\Phi'(\tau) = d\Phi(\tau) = \phi(\tau)d\tau$. So, on the one hand we have

$$\begin{aligned} P(d^2 \leq \tau) &= P\left(\int_{-\infty}^{\infty} [\hat{F}'_n(\tau) - \Phi(\tau)]^2 \phi(\tau) d\tau \leq \tau\right), \\ &= P\left(\int_{-\infty}^{\infty} [\hat{F}'_n(\tau) - \Phi(\tau)]^2 \frac{1}{\sqrt{2\pi}} \exp(-\tau^2/2) d\tau \leq \tau\right), \\ &= P\left(\int_{-\infty}^{\infty} \left[(\hat{F}'_n(\tau))^2 - 2\hat{F}'_n(\tau)\Phi(\tau) + (\Phi(\tau))^2\right] \frac{1}{\sqrt{2\pi}} \exp(-\tau^2/2) d\tau \leq \tau\right), \\ &= P\left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left[(\hat{F}'_n(\tau))^2 - 2\hat{F}'_n(\tau)\Phi(\tau) + (\Phi(\tau))^2\right] \exp(-\tau^2/2) d\tau \leq \tau\right), \quad \tau \in R. \end{aligned}$$

So we started resolving $d^2 = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left[(\hat{F}'_n(\tau))^2 - 2\hat{F}'_n(\tau)\Phi(\tau) + (\Phi(\tau))^2\right] \exp(-\tau^2/2) d\tau$. The quantity d^2 can be written as the sum of three integrals,

$$\begin{aligned} (I) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\hat{F}'_n(\tau))^2 \exp(-\tau^2/2) d\tau \\ (II) &= \sqrt{\frac{2}{\pi}} \int_{-\infty}^{\infty} \hat{F}'_n(\tau)\Phi(\tau) \exp(-\tau^2/2) d\tau \\ (III) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\Phi(\tau))^2 \exp(-\tau^2/2) d\tau. \end{aligned}$$

Resolving the integral (III) we have,

$$\begin{aligned} \Phi(\tau) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\tau} \exp(-t^2/2) dt = \Phi(-\infty, \tau) = \frac{1}{2} + \Phi(\tau) \\ &= \frac{1}{2} + \frac{1}{2} \operatorname{erf}\left(\frac{\tau}{\sqrt{2}}\right) \end{aligned}$$

Thus,

$$\begin{aligned} (III) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\Phi(\tau))^2 \exp(-\tau^2/2) d\tau = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\frac{1}{2} + \frac{1}{2} \operatorname{erf}\left(\frac{\tau}{\sqrt{2}}\right)\right)^2 \exp(-\tau^2/2) d\tau \\ &= 0.33. \end{aligned}$$

Resolving the integral (I),

$$(I) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\hat{F}'_n(\tau))^2 \exp(-\tau^2/2) d\tau$$

$$\begin{aligned}
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\frac{1}{N^m(\alpha, \bar{w})} \sum_{j=1}^{[n/k_n]} \sum_{i=(j-1)k_n+1}^{(j-1)k_n+m_n} I \left(\frac{\sqrt{n}\Delta_i^n X}{\sqrt{\hat{C}_j^n(i)}} \leq \tau \right) I_{(|\Delta_i^n X| \leq \alpha n \bar{w})} \right)^2 \exp(-\tau^2/2) d\tau \\
&= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{N^m(\alpha, \bar{w})} \right)^2 \int_{-\infty}^{\infty} \left(\sum_{j=1}^{[n/k_n]} \sum_{i=(j-1)k_n+1}^{(j-1)k_n+m_n} I \left(\frac{\sqrt{n}\Delta_i^n X}{\sqrt{\hat{C}_j^n(i)}} \leq \tau \right) I_{(|\Delta_i^n X| \leq \alpha n \bar{w})} \right)^2 \exp(-\tau^2/2) d\tau
\end{aligned}$$

and

$$\begin{aligned}
(II) &= \sqrt{\frac{2}{\pi}} \int_{-\infty}^{\infty} \hat{F}'_n(\tau) \Phi(\tau) \exp(-\tau^2/2) d\tau \\
&= \sqrt{\frac{2}{\pi}} \int_{-\infty}^{\infty} \left(\frac{1}{N^m(\alpha, \bar{w})} \sum_{j=1}^{[n/k_n]} \sum_{i=(j-1)k_n+1}^{(j-1)k_n+m_n} I \left(\frac{\sqrt{n}\Delta_i^n X}{\sqrt{\hat{C}_j^n(i)}} \leq \tau \right) I_{(|\Delta_i^n X| \leq \alpha n \bar{w})} \right) \\
&\quad \times \left(\frac{1}{2} + \frac{1}{2} \operatorname{erf} \left(\frac{\tau}{\sqrt{2}} \right) \right) \exp(-\tau^2/2) d\tau \\
&= \sqrt{\frac{2}{\pi}} \frac{1}{N^m(\alpha, \bar{w})} \int_{-\infty}^{\infty} \left(\sum_{j=1}^{[n/k_n]} \sum_{i=(j-1)k_n+1}^{(j-1)k_n+m_n} I \left(\frac{\sqrt{n}\Delta_i^n X}{\sqrt{\hat{C}_j^n(i)}} \leq \tau \right) I_{(|\Delta_i^n X| \leq \alpha n \bar{w})} \right) \\
&\quad \times \left(\frac{1}{2} + \frac{1}{2} \operatorname{erf} \left(\frac{\tau}{\sqrt{2}} \right) \right) \exp(-\tau^2/2) d\tau \\
&= \sqrt{\frac{2}{\pi}} \frac{1}{N^m(\alpha, \bar{w})} \frac{1}{2} \left[\int_{-\infty}^{\infty} \left(\sum_{j=1}^{[n/k_n]} \sum_{i=(j-1)k_n+1}^{(j-1)k_n+m_n} I \left(\frac{\sqrt{n}\Delta_i^n X}{\sqrt{\hat{C}_j^n(i)}} \leq \tau \right) I_{(|\Delta_i^n X| \leq \alpha n \bar{w})} \right) \exp(-\tau^2/2) d\tau \right] \\
&\quad + \int_{-\infty}^{\infty} \left(\sum_{j=1}^{[n/k_n]} \sum_{i=(j-1)k_n+1}^{(j-1)k_n+m_n} I \left(\frac{\sqrt{n}\Delta_i^n X}{\sqrt{\hat{C}_j^n(i)}} \leq \tau \right) I_{(|\Delta_i^n X| \leq \alpha n \bar{w})} \right) \operatorname{erf} \left(\frac{\tau}{\sqrt{2}} \right) \exp(-\tau^2/2) d\tau
\end{aligned}$$

Thus, we can separate (II) as the sum of two integrals given as

$$(*) = \frac{1}{N^m(\alpha, \bar{w})} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\sum_{j=1}^{[n/k_n]} \sum_{i=(j-1)k_n+1}^{(j-1)k_n+m_n} I \left(\frac{\sqrt{n}\Delta_i^n X}{\sqrt{\hat{C}_j^n(i)}} \leq \tau \right) I_{(|\Delta_i^n X| \leq \alpha n \bar{w})} \right) \exp(-\tau^2/2) d\tau$$

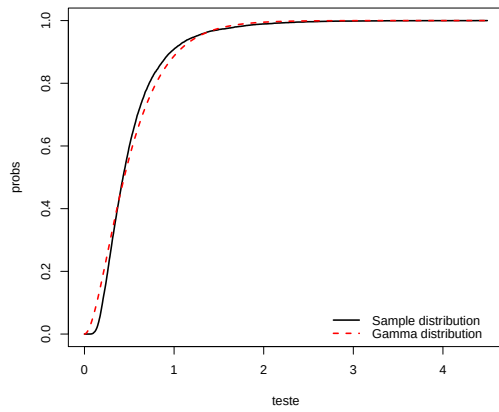
and

$$(**) = \frac{1}{N^m(\alpha, \bar{w})} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\sum_{j=1}^{[n/k_n]} \sum_{i=(j-1)k_n+1}^{(j-1)k_n+m_n} I \left(\frac{\sqrt{n}\Delta_i^n X}{\sqrt{\hat{C}_j^n(i)}} \leq \tau \right) I_{(|\Delta_i^n X| \leq \alpha n \bar{w})} \right) \operatorname{erf} \left(\frac{\tau}{\sqrt{2}} \right) \exp(-\tau^2/2) d\tau.$$

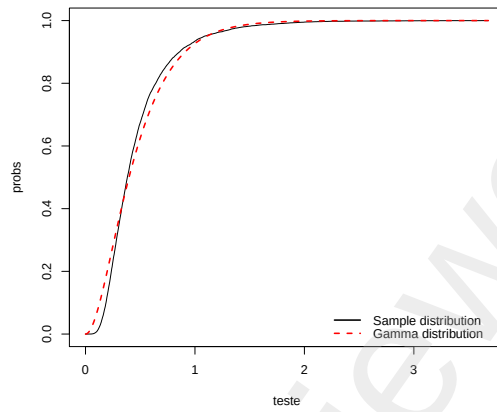
Thus, $(II) = (*) + (**)$. When calculating the cumulative distribution function of our test statistic $TA2$, as can be seen in the sum of (I) (II) and (III) we did not find the closed form and furthermore we did not identify any known expression; this being so, as will be shown in Appendix B; we address our solution to simulation studies.

11 Appendix B

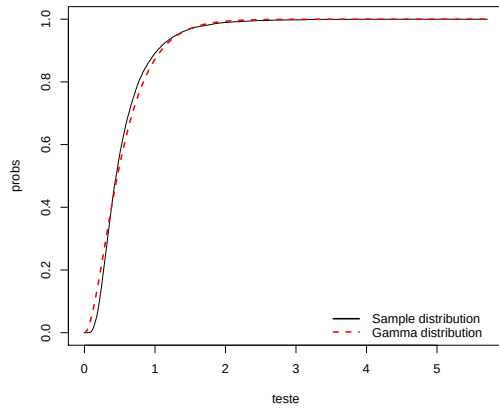
Approximating sampling cumulative distribution function of $TA2$. Nowadays, with the available computational power; if is used of good way, we can resolved or approximate any expression with any function know. For this, is important, have knowledge about expressions analytical and its functional form, in order to obtain our best or at least a good approximation of the distribution



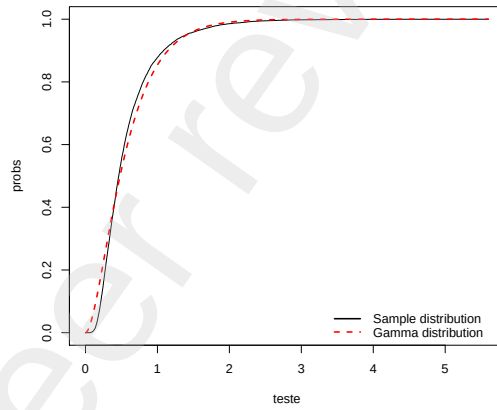
(a) $k_n = 22, n = 500$.



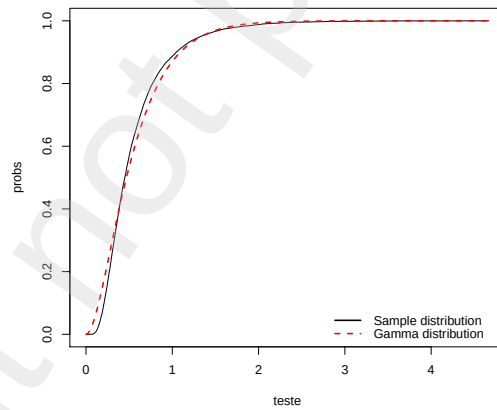
(b) $k_n = 30, n = 1000$.



(c) $k_n = 44, n = 2000$.



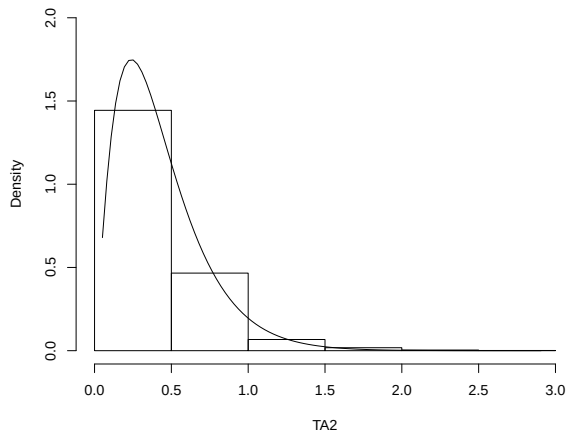
(d) $k_n = 70, n = 5000$.



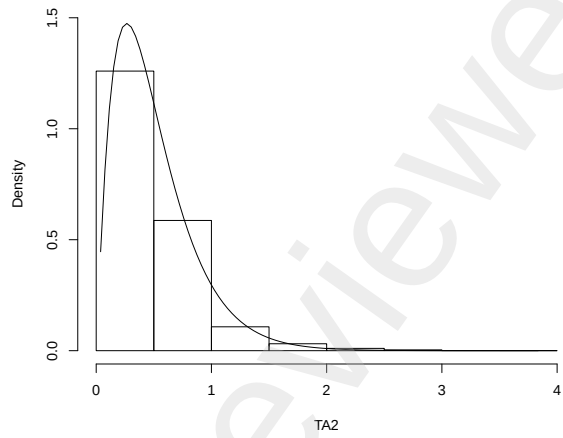
(e) $k_n = 100, n = 10000$.

Figure 19 – Comparison between sampling gamma cumulative distribution function and gamma cumulative distribution function.

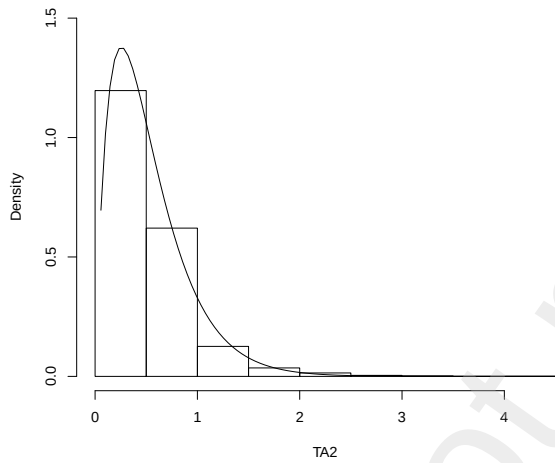
to be found. We observe that the sampling c.d.f. of T_{A2} and gamma c.d.f. are very close. the Figures down show that, we have the excellent approximation with the true density, noticing the good agreement.



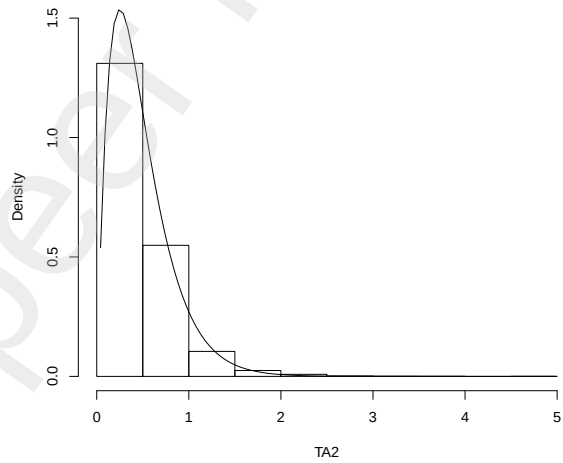
(a) $k_n = 18, n = 500$.



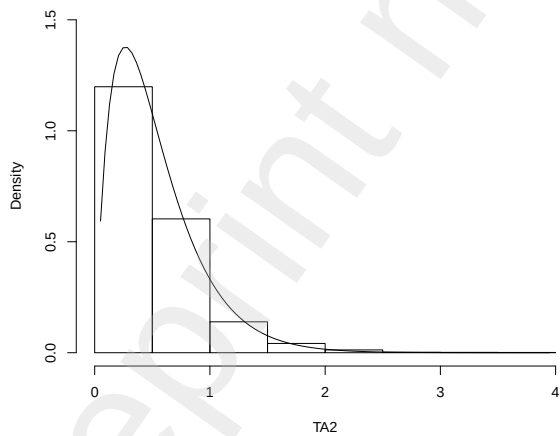
(b) $k_n = 20, n = 500$.



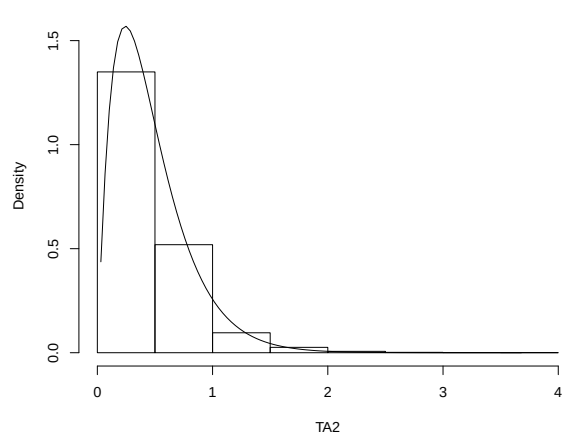
(c) $k_n = 22, n = 500$.



(d) $k_n = 26, n = 1000$.

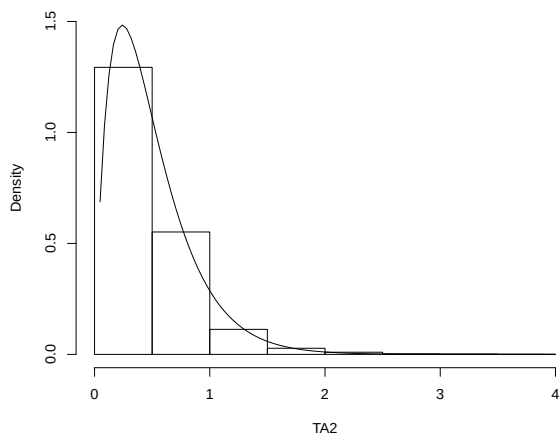


(e) $k_n = 28, n = 1000$.

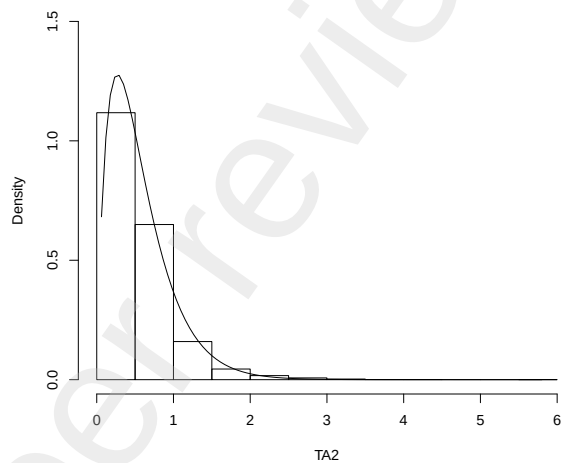


(f) $k_n = 30, n = 1000$.

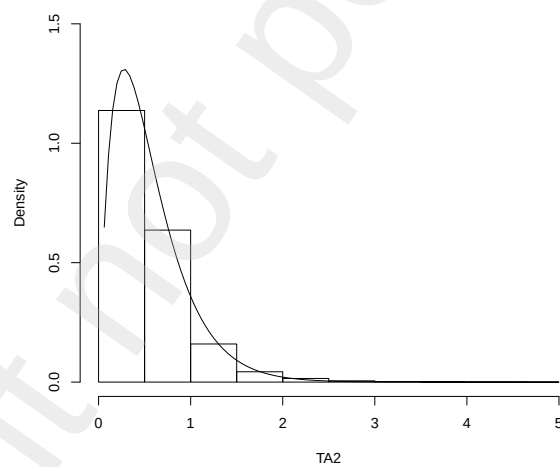
Figure 20 – Histograms of the sample density function for different sizes of k_n and n , compared with the gamma density.



(a) $k_n = 96$, $n = 10000$.



(b) $k_n = 98$, $n = 10000$.



(c) $k_n = 100$, $n = 10000$.

Figure 21 – Histograms of the sample density function for different sizes of k_n and n , compared with the gamma density.